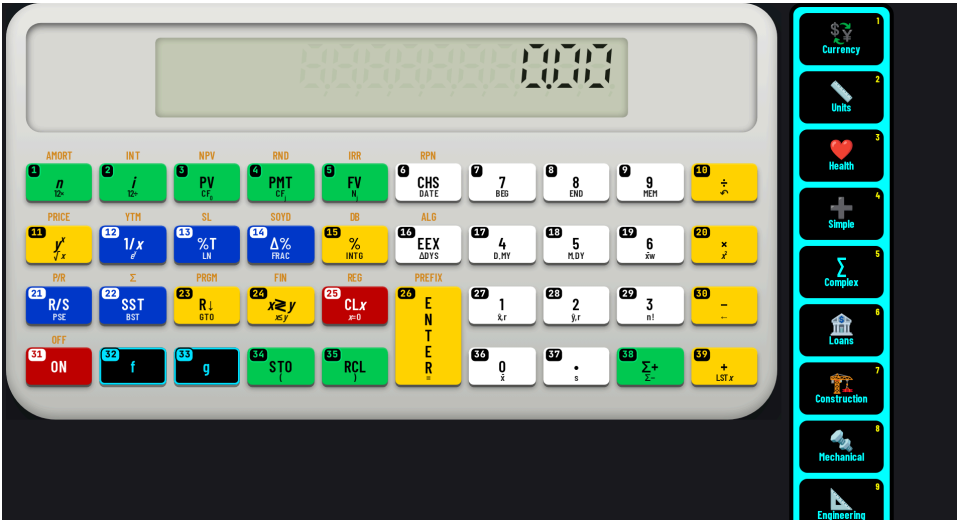
U-45 CALCULATORS

User Manual

Calculator #45 Financial RPN (landscape, platinum)

Financial



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#45 — Financial RPN (landscape, platinum)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example  . Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

RPN financial (ENTER separates numbers; n i PV PMT FV store or solve).

Chapters

1. Getting started
2. Features
3. Numbered key map
4. Key-by-key reference
5. Worked examples
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7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
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12. Appendix C — Glossary

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14. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#45** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#6, #7, #8, #9, #16, #17, #18, #19, #27, #28, #29, #36, #37
op	Operators & execute	#10, #11, #15, #20, #23, #24, #26, #30, #39
fn	Functions	#12, #13, #14, #21, #22
mem	Memory, variables & registers	#1, #2, #3, #4, #5, #34, #35, #38
clr	Clear, delete & power	#25, #31
mod	Shift, modes, menus & navigation	#32, #33

2 · Features

43 features: 26 live on the keys, 9 app/UI features, 8 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Landscape financial RPN with 4-level stack**
2. LIVE **TVM n, i, PV, PMT, FV (enter + key; key alone computes)**
3. LIVE **g 12× / 12÷ monthly conversions**
4. LIVE **BEG / END**
5. LIVE **CFO, CFj, Nj with NPV / IRR**
6. LIVE **AMORT schedule**
7. LIVE **INT simple interest**
8. LIVE **Dates D.MY / M.DY, DATE, ΔDYS**
9. LIVE **Depreciation SL / SOYD / DB**
10. LIVE **Statistics Σ+, x̄, s, ŷ,r, x̂,r, x̄w**
11. LIVE **f + digit sets decimals**
12. LIVE **f CLEAR Σ / FIN / REG / PREFIX**
13. LIVE **%T and Δ%**
14. LIVE **Reciprocal x^{-1} (1/x)**
15. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**
16. LIVE **Polar → rectangular conversion Rec(r,θ) → x, y**
17. LIVE **Rnd — round the value to the displayed precision**
18. LIVE **Percent key (x %, add-on and discount chains)**
19. LIVE **Percent change Δ% between two values**
20. LIVE **Scientific-notation entry ×10^x / EXP / EEX**

21. LIVE Sign change (−) / +/− / CHS
22. LIVE Store a value into memory / a variable (STO, sto►, →, Min)
23. LIVE Recall memory / variables (RCL, MR, recall)
24. LIVE Statistics: data entry, n, $\bar{x}$, σ, s, Σx, Σx^2
25. LIVE Complex numbers: i, arg, conjugate, Re / Im, polar ↔ rectangular
26. LIVE Exchange x ⇄ y / x ⇄ M
27. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
28. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle
29. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
30. APP/UI Show / hide key numbers with one button (remembered between sessions)
31. APP/UI Copy the current display text to the clipboard
32. APP/UI Physical-keyboard shortcuts: digits, + − * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
33. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
34. APP/UI Reference photo viewer and tool-tip name on every key
35. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
36. PLANNED PRICE / YTM bond functions
37. PLANNED Keystroke programming
38. PLANNED Continuous memory snapshot
39. PLANNED Amortisation export (CSV)

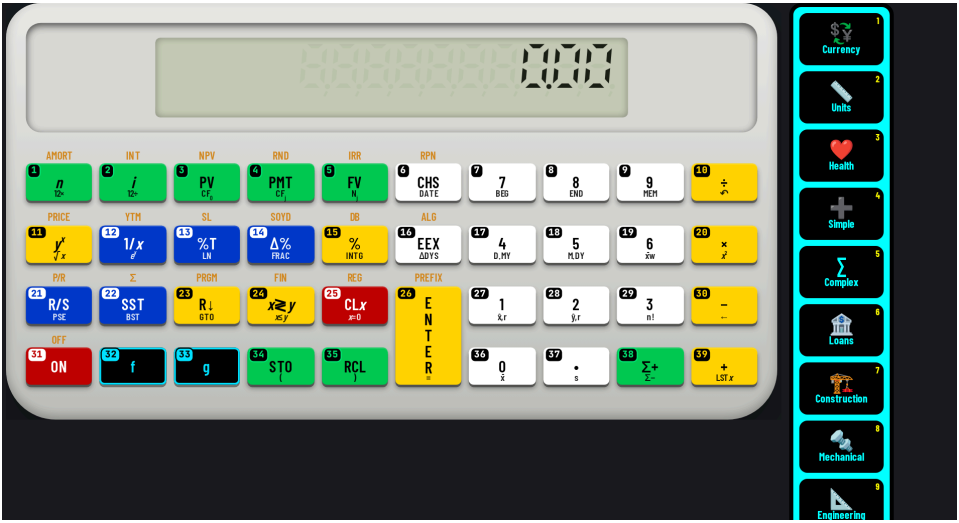
40. PLANNED **Loan comparison**

41. PLANNED **Undo**

42. PLANNED **Speech read-out**

43. PLANNED **Colour-blind key theme**

3 · Numbered key map



Calculator #45 with key numbers shown (Hi-Contrast colours).

Key table

"Shift" = 2nd / f / SHIFT layer; "Alpha" = ALPHA layer; "3rd" = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	n	AMORT	·	·	mem
2	i	INT	·	·	mem
3	PV	NPV	·	·	mem
4	PMT	RND	·	·	mem
5	FV	IRR	·	·	mem
6	CHS	RPN	·	·	num
7	7	·	·	·	num
8	8	·	·	·	num
9	9	·	·	·	num
10	÷	·	·	·	op
11	yx	PRICE	·	·	op
12	1/x	YTM	·	·	fn
13	%T	SL	·	·	fn
14	Δ%	SOYD	·	·	fn
15	%	DB	·	·	op
16	EEX	ALG	·	·	num
17	4	·	·	·	num
18	5	·	·	·	num
19	6	·	·	·	num
20	×	·	·	·	op
21	R/S	P/R	·	·	fn
22	SST	Σ	·	·	fn
23	R↓	PRGM	·	·	op

#	Face	Shift	Alpha	3rd	Group
24	x≷y	FIN	·	·	op
25	CLx	REG	·	·	clr
26	ENTER	PREFIX	·	·	op
27	1	·	·	·	num
28	2	·	·	·	num
29	3	·	·	·	num
30	−	·	·	·	op
31	ON	OFF	·	·	clr
32	f	·	·	·	mod
33	g	·	·	·	mod
34	STO	·	·	·	mem
35	RCL	·	·	·	mem
36	0	·	·	·	num
37	•	·	·	·	num
38	Σ+	·	·	·	mem
39	+	·	·	·	op

4 · Key-by-key reference

Every key on calculator #45, in number order, with every function it carries in this app.

#1 n `Memory, variables & registers`

Primary: n — number of periods (TVM)

shifted (2nd / f / SHIFT layer) (press #32 first): AMORT — AMORT — amortization

Printed under the key: 12×

#2 i `Memory, variables & registers`

Primary: i — interest rate per period (TVM)

shifted (2nd / f / SHIFT layer) (press #32 first): INT — INT — simple interest

Printed under the key: 12÷

#3 PV `Memory, variables & registers`

Primary: PV — present value

shifted (2nd / f / SHIFT layer) (press #32 first): NPV — NPV — net present value of entered cash flows

Printed under the key: CFO

#4 PMT `Memory, variables & registers`

Primary: PMT — payment

shifted (2nd / f / SHIFT layer) (press #32 first): RND — RND — round X to the displayed decimals

Printed under the key: CFj

#5 FV `Memory, variables & registers`

Primary: FV — future value

shifted (2nd / f / SHIFT layer) (press #32 first): IRR — IRR — internal rate of return

Printed under the key: Nj

#6 CHS `Digits & number entry`

Primary: Change sign (+/−) of the number being typed or the current result

Printed under the key: DATE

#7 7 `Digits & number entry`

Primary: Digit 7

Printed under the key: BEG

Keyboard: 7

#8 8 `Digits & number entry`

Primary: Digit 8

Printed under the key: END

Keyboard: 8

#9 9 Digits & number entry

Primary: Digit 9

Printed under the key: MEM

Keyboard: 9

#10 ÷ Operators & execute

Primary: Division ÷

Printed under the key: ↶

Keyboard: /

#11 yx Operators & execute

Primary: y^x — raise Y to the power X (RPN: base first, then exponent)

shifted (2nd / f / SHIFT layer) (press #32 first): PRICE — PRICE — bond price

Printed under the key: √x

Keyboard: ^

#12 1/x Functions

Primary: Reciprocal x^{-1} (1/x)

shifted (2nd / f / SHIFT layer) (press #32 first): YTM — YTM — bond yield

Printed under the key: ex

Keyboard: i

#13 %T Functions

Primary: %T — percent of total

shifted (2nd / f / SHIFT layer) (press #32 first): SL — SL — straight-line depreciation

Printed under the key: LN

#14 Δ% Functions

Primary: Δ% — percent change

shifted (2nd / f / SHIFT layer) (press #32 first): SOYD — SOYD — sum-of-years depreciation

Printed under the key: FRAC

#15 % Operators & execute

Primary: Percent %

shifted (2nd / f / SHIFT layer) (press #32 first): DB — DB — declining-balance depreciation

Printed under the key: INTG

Keyboard: %

#16 EEX Digits & number entry

Primary: EEX — enter a power-of-ten exponent for the number in X

Printed under the key: ΔDYS

Keyboard: e

#17 **4** Digits & number entry

Primary: Digit 4

Printed under the key: D.MY

Keyboard: 4

#18 **5** Digits & number entry

Primary: Digit 5

Printed under the key: M.DY

Keyboard: 5

#19 **6** Digits & number entry

Primary: Digit 6

Printed under the key: x̄w

Keyboard: 6

#20 **×** Operators & execute

Primary: Multiplication ×

Printed under the key: x^2

Keyboard: * x

#21 R/S Functions

Primary: R/S — run/stop

shifted (2nd / f / SHIFT layer) (press #32 first): P/R — P/R — program mode

Printed under the key: PSE

#22 SST Functions

Primary: SST — single step

shifted (2nd / f / SHIFT layer) (press #32 first): Σ — CLEAR Σ — clear statistics registers

Printed under the key: BST

#23 R↓ Operators & execute

Primary: R↓ — roll the stack down

shifted (2nd / f / SHIFT layer) (press #32 first): PRGM — Shows the message "Programs n/a" (feature not provided in this app)

Printed under the key: GTO

#24 x≷y Operators & execute

Primary: x≷y — exchange X and Y

shifted (2nd / f / SHIFT layer) (press #32 first): FIN — CLEAR FIN — clear the financial (TVM) registers

Printed under the key: x≤y

#25 CLx `Clear, delete & power`

Primary: CLx — clear X

shifted (2nd / f / SHIFT layer) (press #32 first): REG — CLEAR REG — clear all registers

Printed under the key: x=0

#26 ENTER `Operators & execute`

Primary: ENTER — separate two numbers (push X to Y)

shifted (2nd / f / SHIFT layer) (press #32 first): PREFIX — CLEAR PREFIX — cancel a pending f / g / STO / RCL prefix

Printed under the key: =

Keyboard: `Enter`

#27 1 `Digits & number entry`

Primary: Digit 1

Printed under the key: x̂,r

Keyboard: `1`

#28 2 `Digits & number entry`

Primary: Digit 2

Printed under the key: ŷ,r

Keyboard: `2`

#29 3 `Digits & number entry`

Primary: Digit 3

Printed under the key: n!

Keyboard: 3

#30 – `Operators & execute`

Primary: Subtraction –

Printed under the key: ←

Keyboard: -

#31 ON `Clear, delete & power`

Primary: ON/OFF — power toggle

shifted (2nd / f / SHIFT layer) (press #32 first): OFF — ON/OFF — power toggle

#32 f `Shift, modes, menus & navigation`

Primary: f — gold prefix: next key does its upper function

#33 g `Shift, modes, menus & navigation`

Primary: g — blue prefix: next key does its lower function

#34 STO `Memory, variables & registers`

Primary: STO — store the value into a memory / variable (press the target next)

Printed under the key: (

#35 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

Printed under the key:)

#36 0 Digits & number entry

Primary: Digit 0

Printed under the key: x̄

Keyboard: 0

#37 • Digits & number entry

Primary: Decimal point. Pressing it first gives "0."; a second point in the same number is ignored

Printed under the key: s

Keyboard: . ,

#38 Σ+ Memory, variables & registers

Primary: Σ+ — enter a statistics data point

Printed under the key: Σ−

#39 + Operators & execute

Primary: Addition +

Printed under the key: LSTx

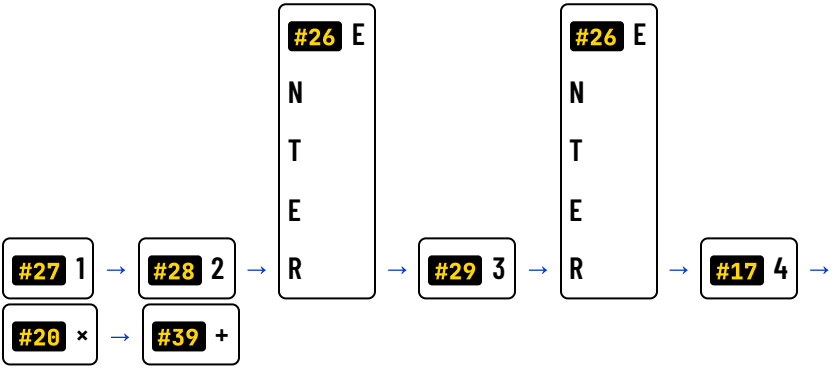
Keyboard: +

5 · Worked examples

15 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

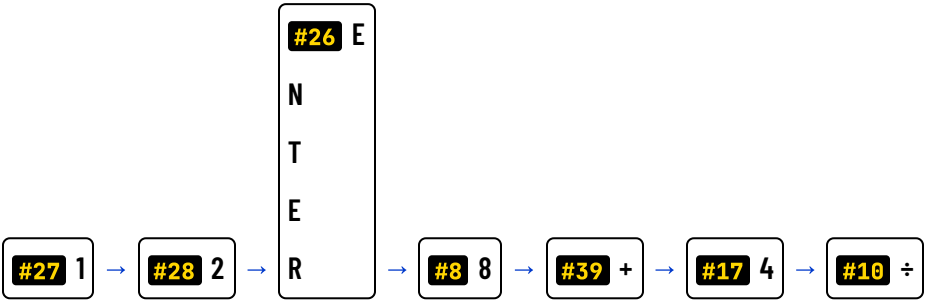


Display shows 24.00 **— expected 24. ✔ verified on the app.**

RPN needs no precedence rules: the product is formed first because you key it first, then + adds it to the number waiting on the stack.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

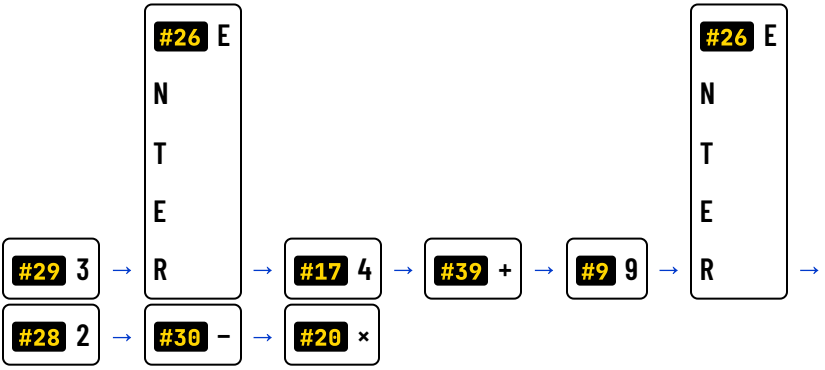


Display shows 5.00 **— expected 5. ✔ verified on the app.**

In RPN the brackets disappear — add first, then divide.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

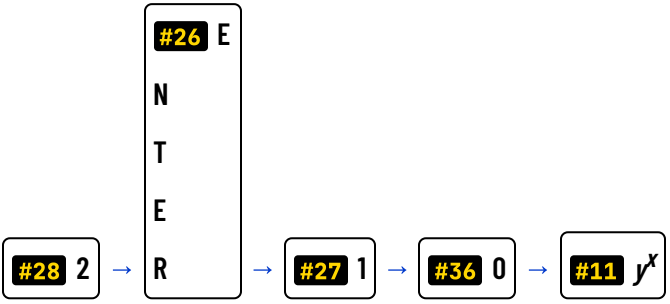


Display shows 49.00 **— expected 49. ✔ verified on the app.**

Each bracket becomes its own stack result; the final × combines the two.

Example 5.4 — Powers

Calculate 2^10. Press Reset first.



Display shows 1,024.00 **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.5 — Reciprocals

Calculate 1 ÷ 4 (x^{-1}). Press Reset first.

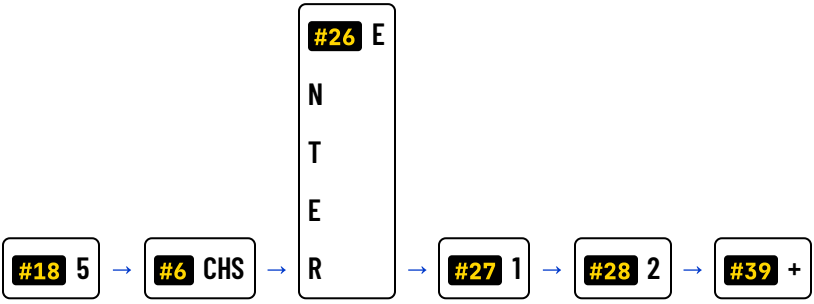


Display shows 0.25 **— expected 0.25. ✔ verified on the app.**

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.6 — Negative numbers

Calculate −5 + 12. Press Reset first.

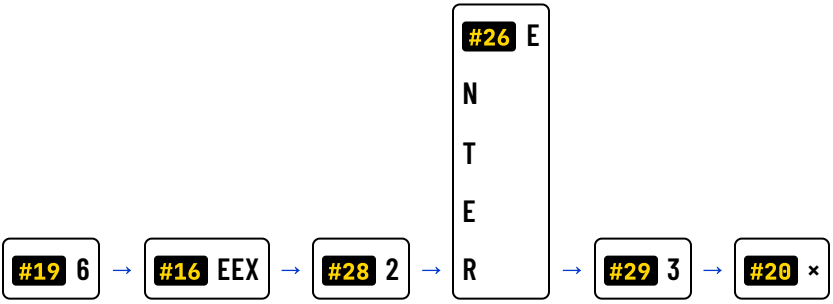


Display shows 7.00 **— expected 7. ✔ verified on the app.**

Use the sign key for a negative number — not the subtraction key.

Example 5.7 — Scientific notation

Calculate 6×10^2 × 3. Press Reset first.

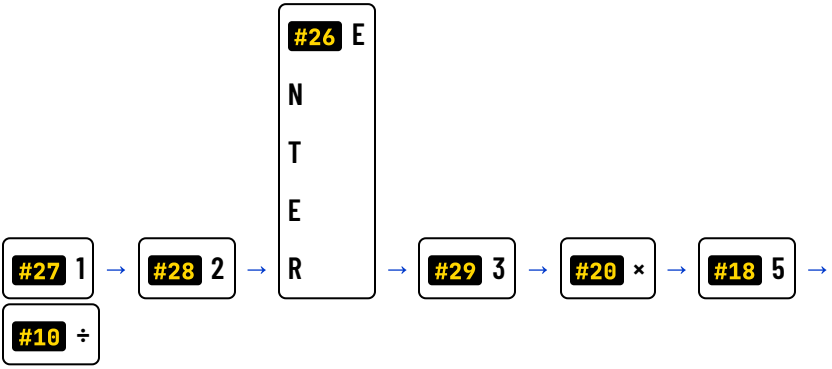


Display shows 1,800.00 **— expected 1800. ✔ verified on the app.**

The exponent key enters ×10ⁿ without typing × 10 ^.

Example 5.8 — Continue from the answer

Calculate 12 × 3 = , then Ans ÷ 5. Press Reset first.

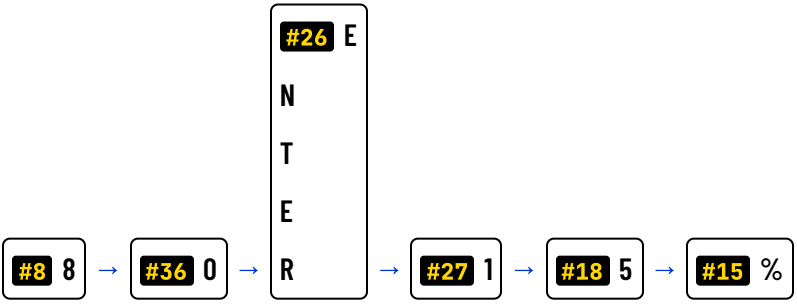


Display shows 7.20 **— expected 7.2. ✔ verified on the app.**

The result simply stays in X, ready for the next operation.

Example 5.9 — Percent of an amount

Calculate 15 % of 80. Press Reset first.

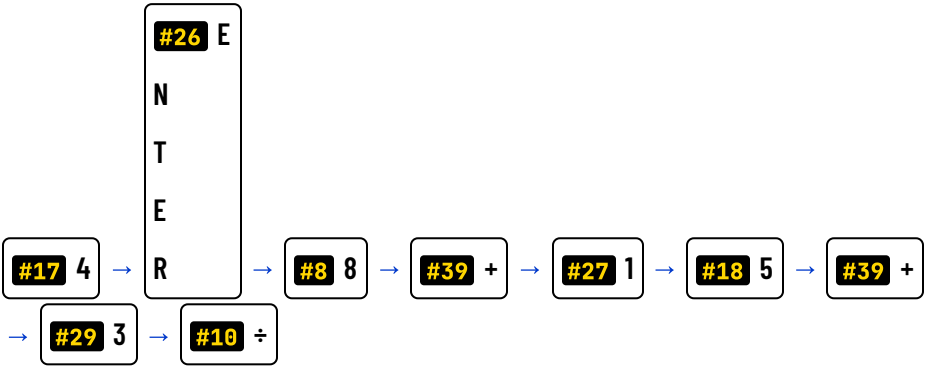


Display shows 12.00 **— expected 12. ✔ verified on the app.**

The percent key turns the second number into a percentage of the first.

Example 5.10 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

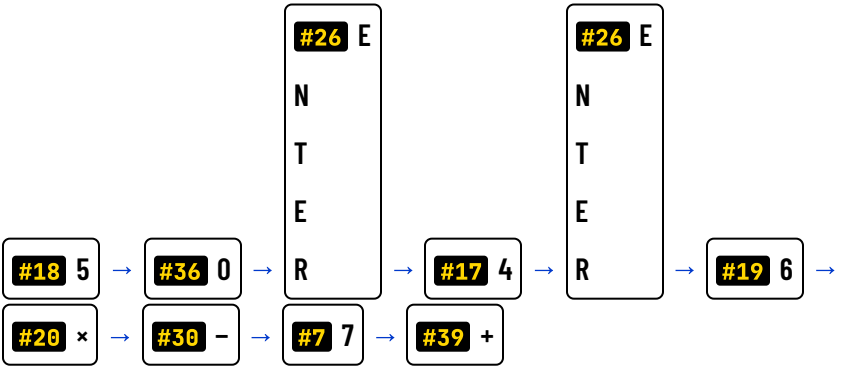


Display shows 9.00 **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.11 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

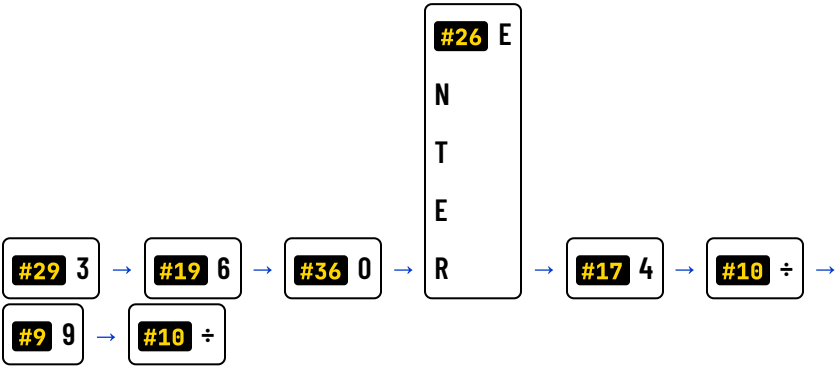


Display shows 33.00 **— expected 33. ✔ verified on the app.**

Key the product first (it waits on the stack), subtract it, then add the last value.

Example 5.12 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

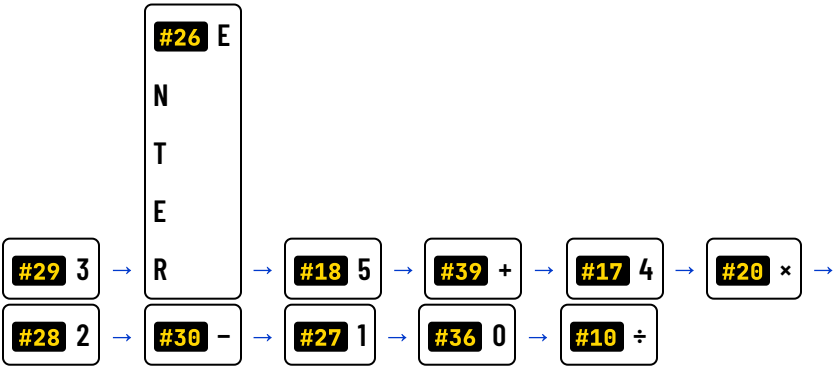


Display shows 10.00 **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.13 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.



Display shows 3.00 **— expected 3. ✔ verified on the app.**

Work from the innermost bracket outwards — each step leaves its result in X.

Example 5.14 — Loan payment (TVM)

Calculate Loan 100000, 6 % a year, 360 monthly payments → PMT. Press Reset first.

#29 3 → #19 6 → #36 0 → #1 *n* → #19 6 → #26 ENTER → #27 1 → #28 2 → #10 ÷ → #2 *i* → #27 1 → #36 0 → #36 0 → #36 0 → #36 0 → #36 0 → #3 PV → #36 0 → #5 FV → #4 PMT

Display shows `-599.55` **— expected -599.55. ✔ verified on the app.**

Money paid out is negative: a positive loan (PV) produces a negative payment.

Example 5.15 — Savings growth (TVM)

Calculate Deposit 1000 for 10 years at 5 % a year (yearly) → FV. Press Reset first.

#27 1 → #36 0 → #1 *n* → #18 5 → #2 *i* → #27 1 → #36 0 → #36 0 → #36 0 → #36 0 → #6 CHS → #3 PV → #36 0 → #4 PMT → #5 FV

Display shows `1,628.89` **— expected 1628.89. ✔ verified on the app.**

The deposit is entered as negative (money out), so the future value comes back positive.

6 · Tricks & hacks

Hack 1: No equals key

RPN works number, number, operator: 3 ENTER 4 + = 7. Results stay on the stack for the next step.

Hack 2: Chain without ENTER

After an operation, just type the next number and operator: 3 ENTER 4 + 5 × gives (3+4)×5 = 35.

Hack 3: Swap to fix order

If you entered numbers in the wrong order for − ÷ or y^x, press x≷y before the operator.

Hack 4: LASTx to reuse

LASTx brings back the X value of the previous operation — useful for a × a.

Hack 5: Stack as memory

Up to four values are kept; R↓ rolls through them.

Hack 6: Constant via stack top

On 4-level stacks the T register copies down, so a constant multiplier can be reused by pressing × repeatedly.

Hack 7: Sign convention

Money you receive is positive, money you pay is negative. A loan: PV = +100000, PMT comes out negative.

Hack 8: Clear TVM first

Always clear the TVM registers before a new problem so old values don't leak in.

Hack 9: Check BEG/END

Payments at the beginning of each period (leases, rent) need BEG mode.

Hack 10: Monthly rates

Convert yearly figures: n = years × 12, i = yearly % ÷ 12 (on models with periodic i).

Hack 11: Worksheets

Use ↑ ↓ to move through a worksheet, ENTER to store, CPT to compute.

Hack 12: Hidden functions on this model

17 extra functions sit on shifted layers. Press the layer key once, then the key: for example #32 then #1 → AMORT; #32 then #2 → INT; #32 then #3 → NPV; #32 then #4 → RND.

Hack 13: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #39.

Hack 14: Type faster on a computer

20 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 15: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 16: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 17: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
7	#7	7	Digit 7
8	#8	8	Digit 8
9	#9	9	Digit 9
/	#10	÷	Division ÷
^	#11	yx	y^x — raise Y to the power X (RPN: base first, then exponent)
i	#12	1/x	Reciprocal x^{-1} (1/x)
%	#15	%	Percent %
e	#16	EEX	EEX — enter a power-of-ten exponent for the number in X
4	#17	4	Digit 4
5	#18	5	Digit 5
6	#19	6	Digit 6
* x	#20	×	Multiplication ×
Enter	#26	ENTER	ENTER — separate two numbers (push X to Y)
1	#27	1	Digit 1
2	#28	2	Digit 2
3	#29	3	Digit 3
-	#30	−	Subtraction −
0	#36	0	Digit 0
. ,	#37	•	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored
+	#39	+	Addition +

App shortcuts

“# On / # Off”	**show / hide key numbers**
“Classic / Hi-Contrast”	**switch key colours**
“Copy”	**copy display text**
“Fit / 1:1”	**scale to screen / design size**
“Reset”	**restart this calculator**
Jump list	**switch to another of the 45 calculators**

8 · Common mistakes

Mistake 1: Pressing ENTER after an operation

ENTER after + duplicates the result. Usually you want to type the next number directly.

Mistake 2: Wrong order for – and ÷

Y – X and Y ÷ X: enter the first operand first. 10 ENTER 4 – = 6.

Mistake 3: CHS vs –

Make a number negative with CHS/+/–, not with –.

Mistake 4: Prefix left active

If f / g / SHIFT is shown, the next key is shifted. Clear it with the prefix-clear function.

Mistake 5: Same sign on PV and FV

If PV and FV have the same sign, there is no interest rate solution — one must be negative.

Mistake 6: Leftover values

Not clearing TVM is the most common cause of wrong payments.

Mistake 7: Percent vs decimal

Enter rates as percentages (6 for 6 %), not 0.06.

Mistake 8: Chain logic

Financial models may evaluate left-to-right in chain mode: 2 + 3 × 4 = 20. Use brackets or switch to AOS.

Mistake 9: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 10: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (97)

Work each one on calculator #45, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 2.5 + 4 × 1.5 = _____
7. 1000 + 25 × 4 = _____
8. 19 + 21 × 3 = _____
9. 64 + 8 × 8 = _____
10. 0.5 + 0.25 × 8 = _____
11. (45 + 15) ÷ 6 = _____
12. (9 + 7) ÷ 2 = _____
13. (100 + 60) ÷ 8 = _____
14. (33 + 17) ÷ 5 = _____
15. (7.5 + 2.5) ÷ 4 = _____
16. (120 + 80) ÷ 16 = _____

17. (1 + 2) ÷ 4 = _____

18. (250 + 350) ÷ 12 = _____

19. (5 + 6) × (10 − 3) = _____

20. (12 + 3) × (8 − 5) = _____

21. (1.5 + 2.5) × (7 − 4) = _____

22. (20 + 30) × (15 − 5) = _____

23. (9 + 1) × (100 − 1) = _____

24. (11 + 13) × (17 − 9) = _____

25. 3^5 = _____

26. 1.05^3 = _____

27. 5^4 = _____

28. 7^3 = _____

29. 1.5^2 = _____

30. 10^6 = _____

31. 2^0.5 = _____

32. 1 ÷ 8 (x^{-1}) = _____

33. 1 ÷ 0.25 (x^{-1}) = _____

34. 1 ÷ 16 (x^{-1}) = _____

35. 1 ÷ 5 (x^{-1}) = _____

36. 1 ÷ 40 (x^{-1}) = _____

37. $1 \div 0.2\ (x^{-1}) =$ _____

38. $-20 + 7 =$ _____

39. $-3 + 3 =$ _____

40. $-15.5 + 4 =$ _____

41. $-100 + 250 =$ _____

42. $-8 + 1 =$ _____

43. $2\times10^{3} \times 4 =$ _____

44. $1.5\times10^{4} \times 2 =$ _____

45. $3\times10^{5} \times 3 =$ _____

46. $4.5\times10^{2} \times 2 =$ _____

47. $100 \times 4 =$, then Ans $\div 2 =$ _____

48. $15 \times 6 =$, then Ans $\div 9 =$ _____

49. $7.5 \times 8 =$, then Ans $\div 3 =$ _____

50. $250 \times 12 =$, then Ans $\div 100 =$ _____

51. 8 % of 250 = _____

52. 25 % of 360 = _____

53. 12.5 % of 40 = _____

54. 50 % of 75 = _____

55. 10 % of 450 = _____

56. 20 % of 125 = _____

57. 5 % of 640 = _____

58. 75 % of 48 = _____

59. 30 % of 90 = _____

60. (12 + 15 + 18) ÷ 3 = _____

61. (70 + 85 + 91) ÷ 3 = _____

62. (2.5 + 3.5 + 6) ÷ 3 = _____

63. (100 + 200 + 600) ÷ 3 = _____

64. 100 − 3 × 12 + 9 = _____

65. 18 − 2 × 5 + 40 = _____

66. 7.5 − 1.5 × 3 + 2 = _____

67. 64 − 8 × 4 + 10 = _____

68. 300 − 25 × 4 + 55 = _____

69. 9 − 3 × 3 + 1 = _____

70. 45 − 5 × 5 + 15 = _____

71. 1000 ÷ 8 ÷ 5 = _____

72. 96 ÷ 2 ÷ 3 = _____

73. 7200 ÷ 60 ÷ 60 = _____

74. 500 ÷ 4 ÷ 25 = _____

75. 81 ÷ 3 ÷ 3 = _____

76. 1.5 ÷ 0.5 ÷ 2 = _____

77. ((10 + 2) × 3 − 6) ÷ 6 = _____

78. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

79. ((20 + 30) × 2 − 25) ÷ 3 = _____

80. ((7 + 8) × 9 − 15) ÷ 12 = _____

81. Loan 25000, 4.5 % a year, 60 monthly payments → PMT = _____

82. Deposit 5000 for 20 years at 3 % a year (yearly) → FV = _____

83. 12 + 3 × 4 = _____

84. (12 + 8) ÷ 4 = _____

85. (3 + 4) × (9 − 2) = _____

86. 2^10 = _____

87. 1 ÷ 4 (x^{-1}) = _____

88. −5 + 12 = _____

89. 6×10^2 × 3 = _____

90. 12 × 3 = , then Ans ÷ 5 = _____

91. 15 % of 80 = _____

92. (4 + 8 + 15) ÷ 3 = _____

93. 50 − 4 × 6 + 7 = _____

94. 360 ÷ 4 ÷ 9 = _____

95. ((3 + 5) × 4 − 2) ÷ 10 = _____

96. Loan 100000, 6 % a year, 360 monthly payments → PMT = _____

97. Deposit 1000 for 10 years at 5 % a year (yearly) → FV = _____

Set B — Key-finding drills (46)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “n” (n)? _____
2. Which key is “i” (i)? _____
3. Which key is “PV” (PV)? _____
4. Which key is “PMT” (PMT)? _____
5. Which key is “FV” (FV)? _____
6. Which key is “CHS” (Change sign)? _____
7. Which key is “÷” (Division ÷)? _____
8. Which key is “yx” (y^x)? _____
9. Which key is “1/x” (Reciprocal x^{-1})? _____
10. Which key is “%T” (%T)? _____
11. Which key is “Δ%” (Δ%)? _____
12. Which key is “%” (Percent %)? _____
13. Which key is “EEX” (EEX)? _____
14. Which key is “×” (Multiplication ×)? _____
15. Which key is “R/S” (R/S)? _____
16. Which key is “SST” (SST)? _____
17. Which key is “R↓” (R↓)? _____

18. Which key is "x≷y" (x≷y)? _____

19. Which key is "CLx" (CLx)? _____

20. Which key is "ENTER" (ENTER)? _____

21. Which key is "–" (Subtraction –)? _____

22. Which key is "ON" (ON/OFF)? _____

23. Which key is "f" (f)? _____

24. Which key is "g" (g)? _____

25. Which key is "STO" (STO)? _____

26. Which key is "RCL" (RCL)? _____

27. Which key is "•" (Decimal point. Pressing it first gives "0."; a second point in the same number is ignored)? _____

28. Which key is "Σ+" (Σ+)? _____

29. Which key is "+" (Addition +)? _____

30. How do you reach "AMORT"? _____

31. How do you reach "INT"? _____

32. How do you reach "NPV"? _____

33. How do you reach "RND"? _____

34. How do you reach "IRR"? _____

35. How do you reach "PRICE"? _____

36. How do you reach "YTM"? _____

37. How do you reach "SL"? _____

38. How do you reach “SOYD”? _____

39. How do you reach “DB”? _____

40. How do you reach “P/R”? _____

41. How do you reach “Σ”? _____

42. How do you reach “PRGM”? _____

43. How do you reach “FIN”? _____

44. How do you reach “REG”? _____

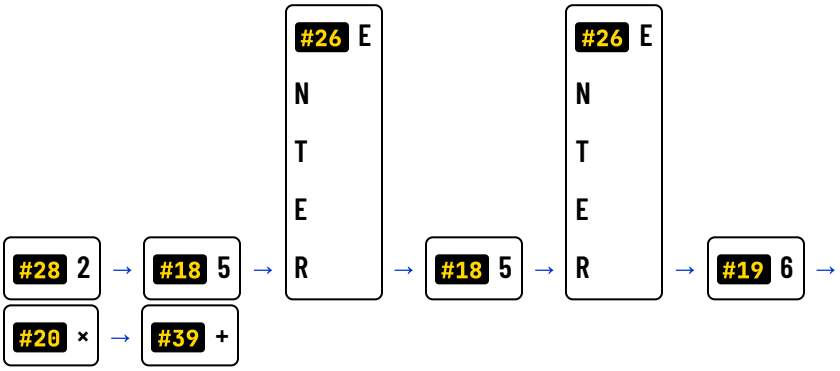
45. How do you reach “PREFIX”? _____

46. How do you reach “OFF”? _____

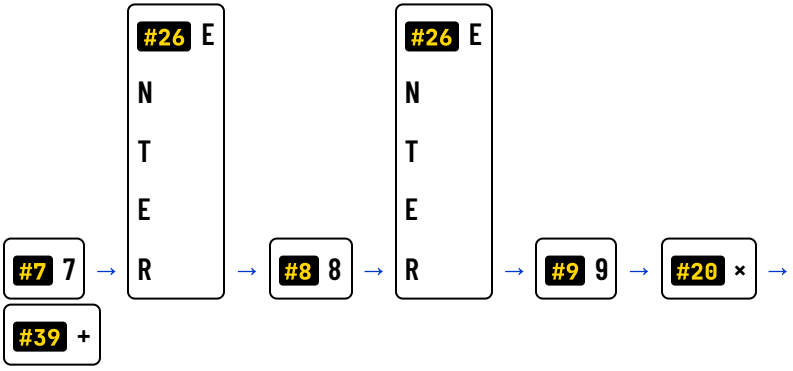
Appendix A — Drill answers

Set A

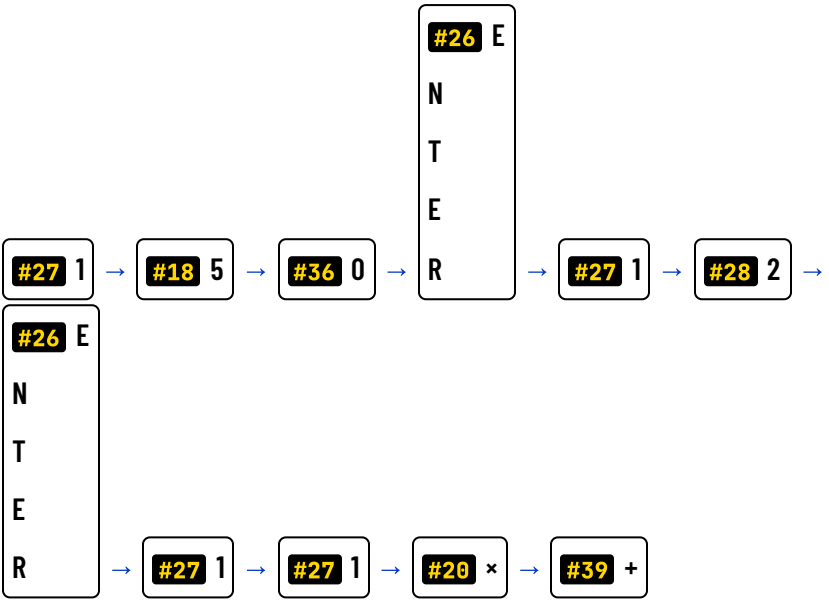
1. 55 (display 55.00)



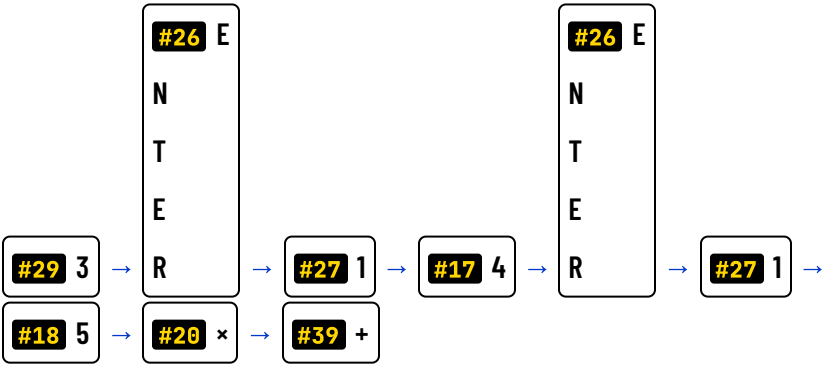
2. 79 (display 79.00)



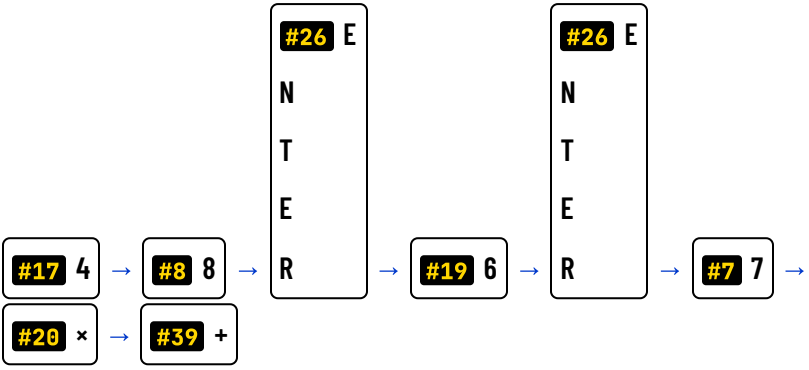
3. 282 (display 282.00)



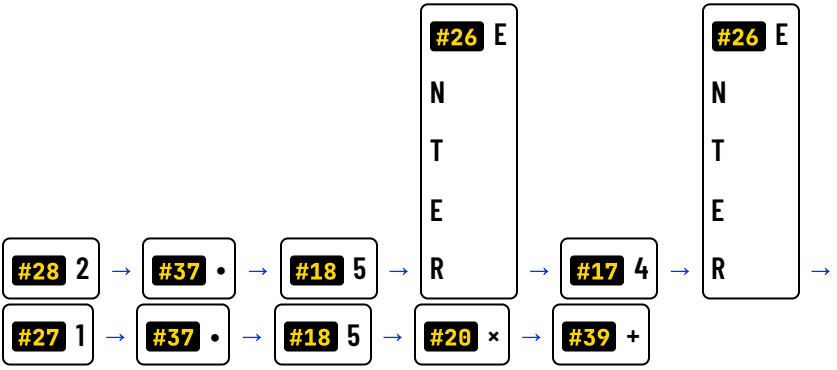
4. 213 (display 213.00)



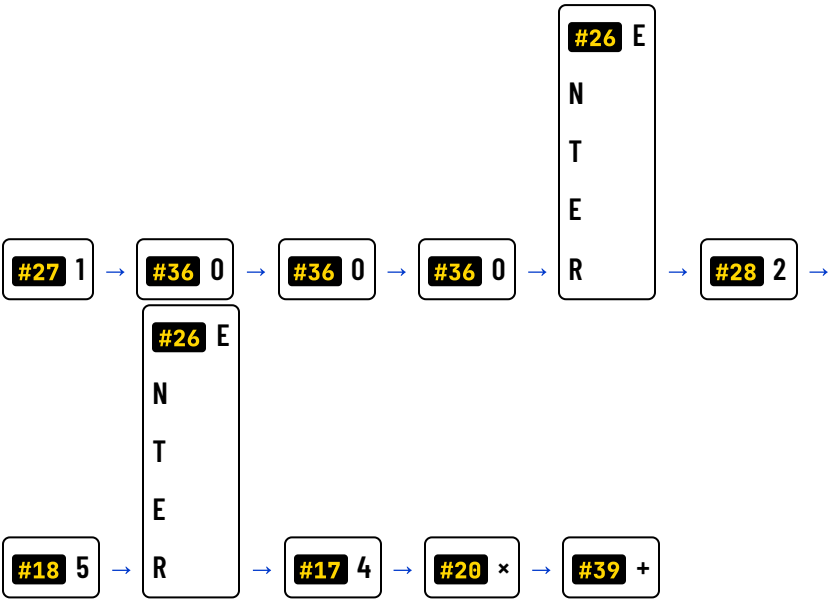
5. 90 (display 90.00)



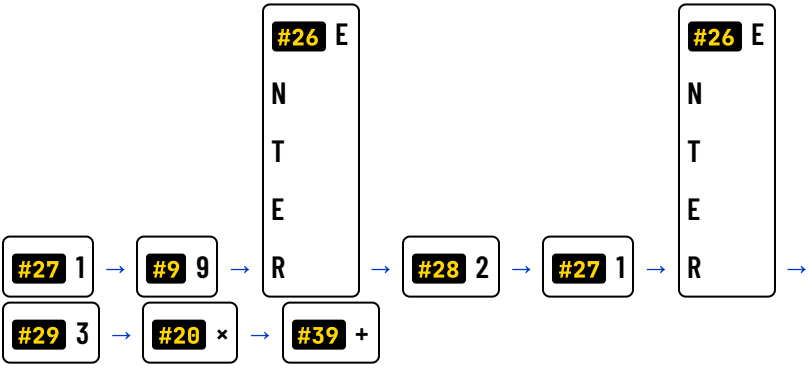
6. 8.5 (display 8.50)



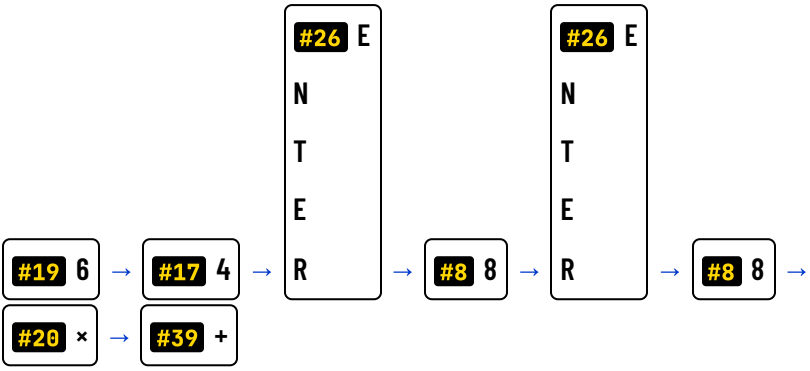
7. 1100 (display 1,100.00)



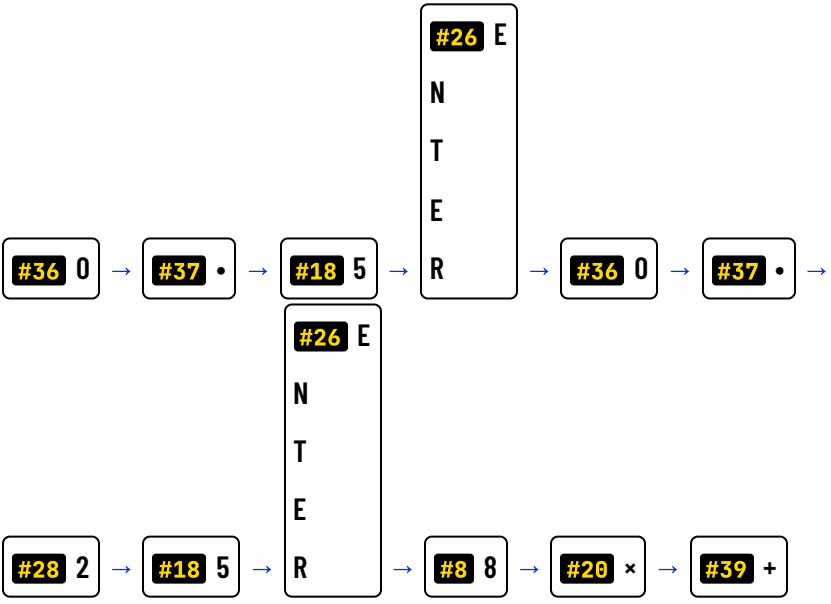
8. 82 (display 82.00)



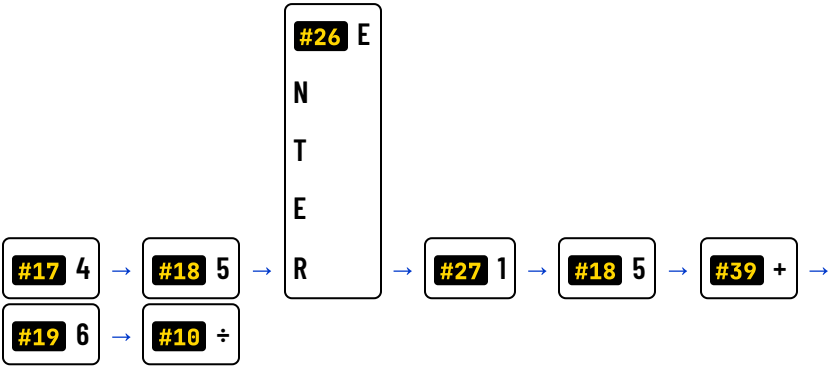
9. 128 (display 128.00)



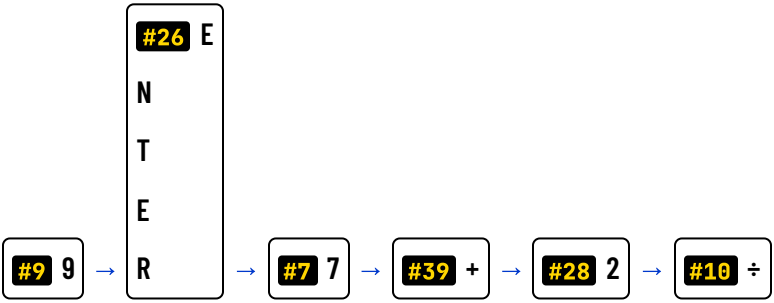
10. 2.5 (display 2.50)



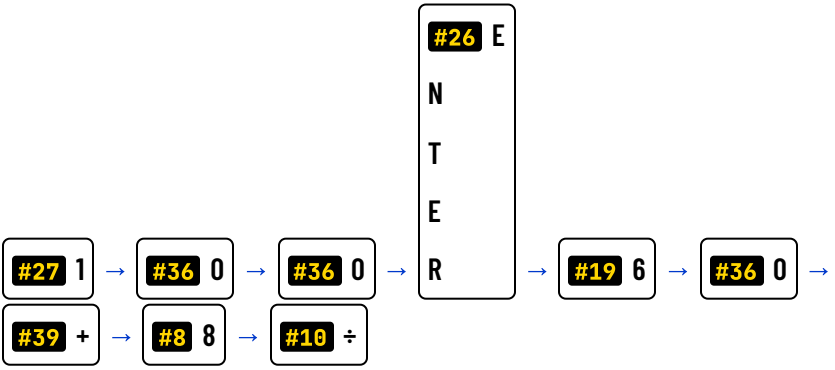
11. 10 (display 10.00)



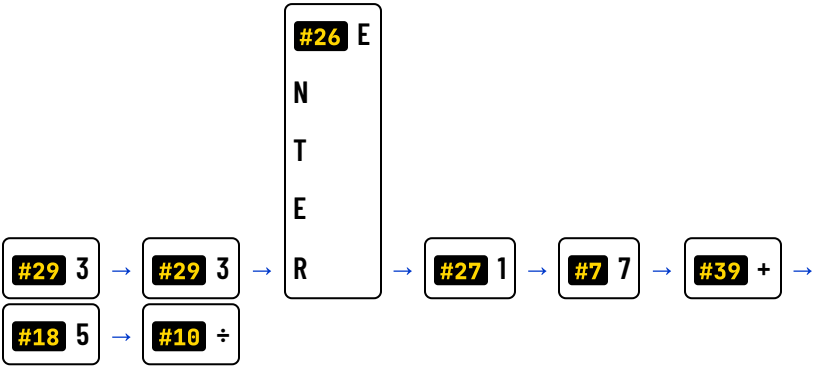
12. 8 (display 8.00)



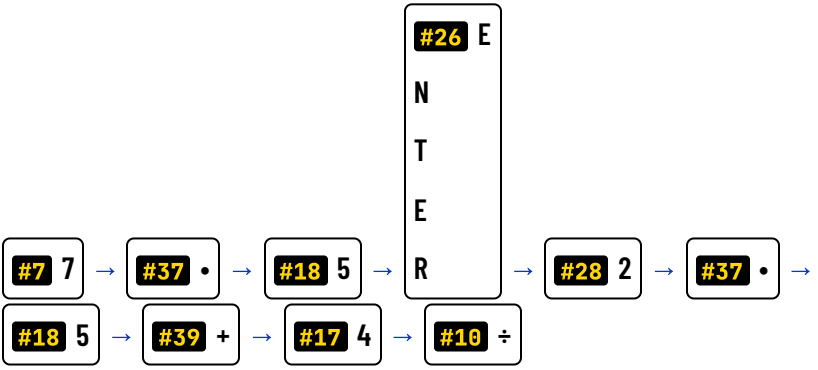
13. 20 (display 20.00)



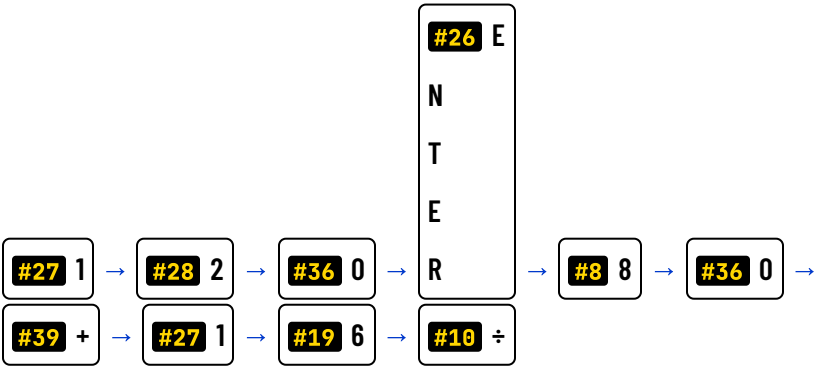
14. 10 (display 10.00)



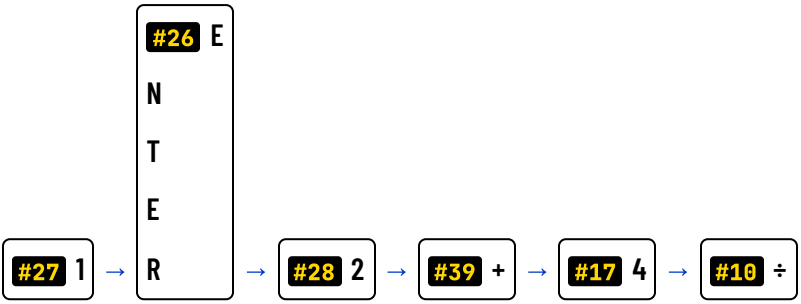
15. 2.5 (display 2.50)



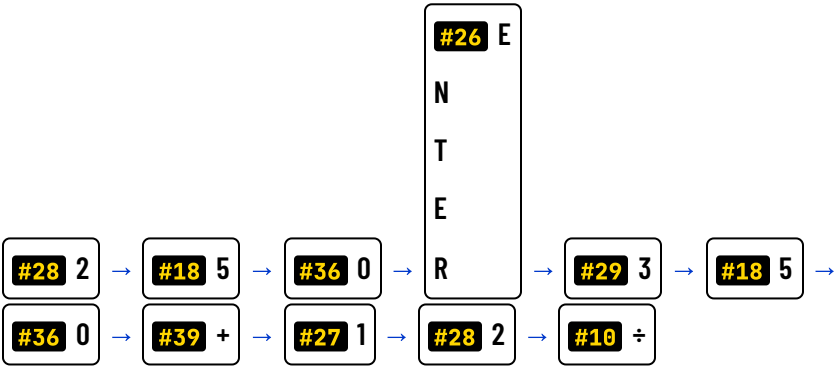
16. 12.5 (display 12.50)



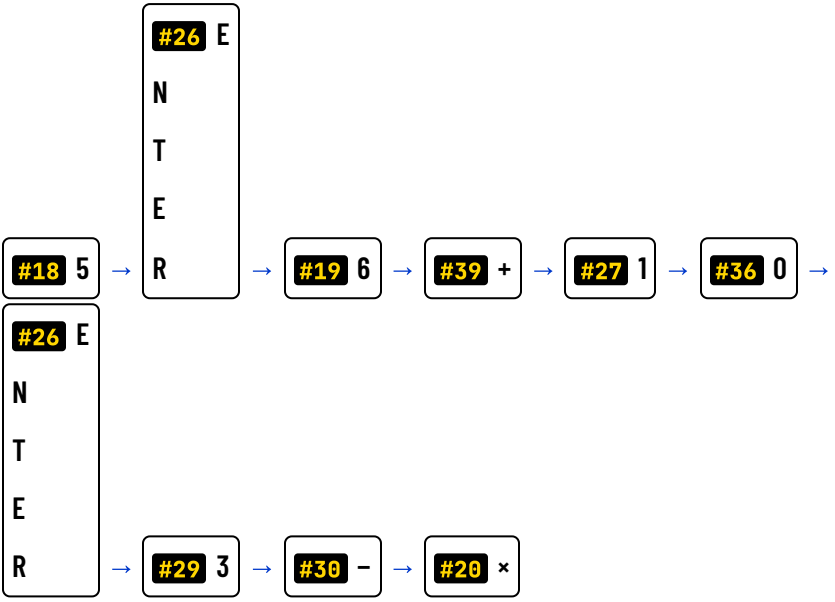
17. 0.75 (display 0.75)



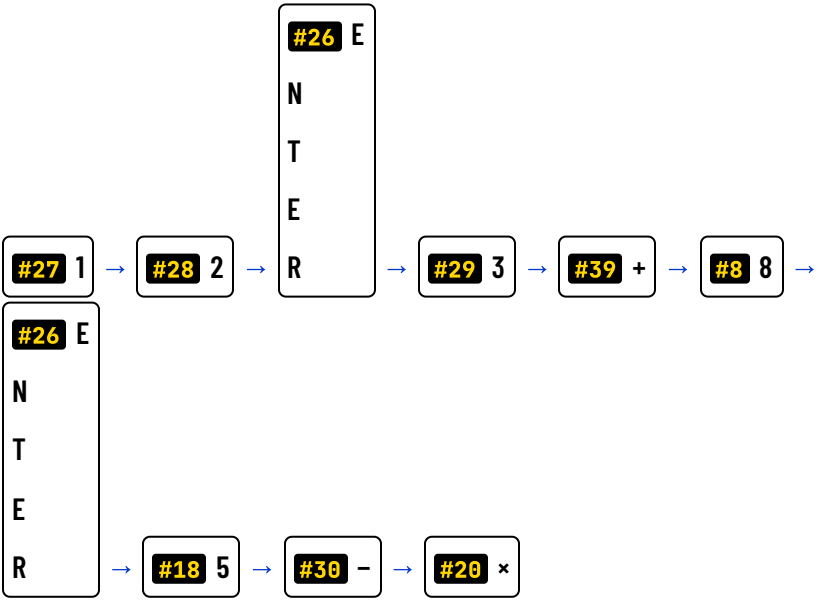
18. 50 (display 50.00)



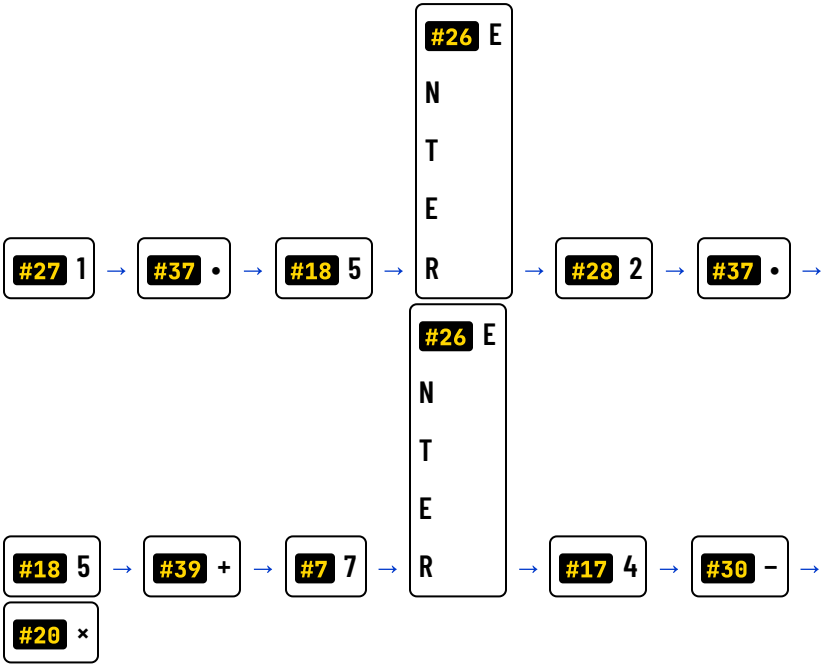
19. 77 (display 77.00)



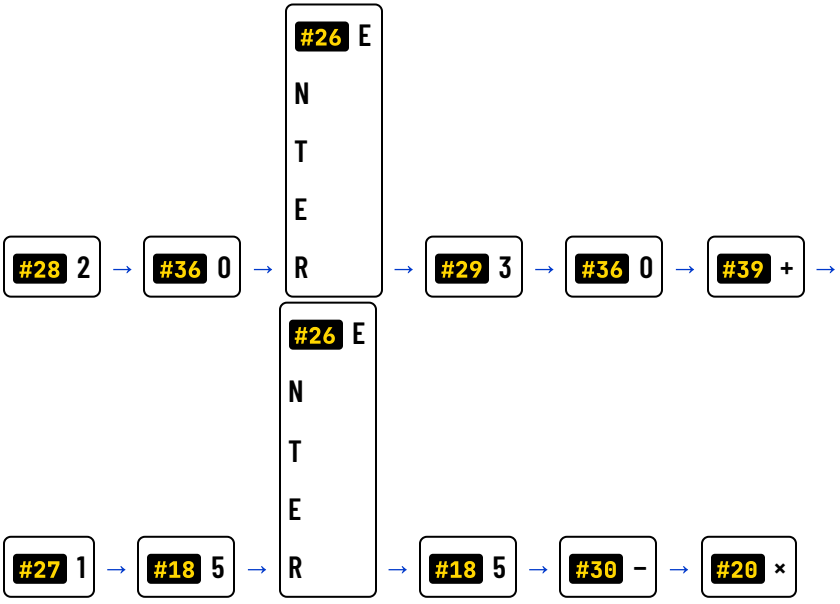
20. 45 (display 45.00)



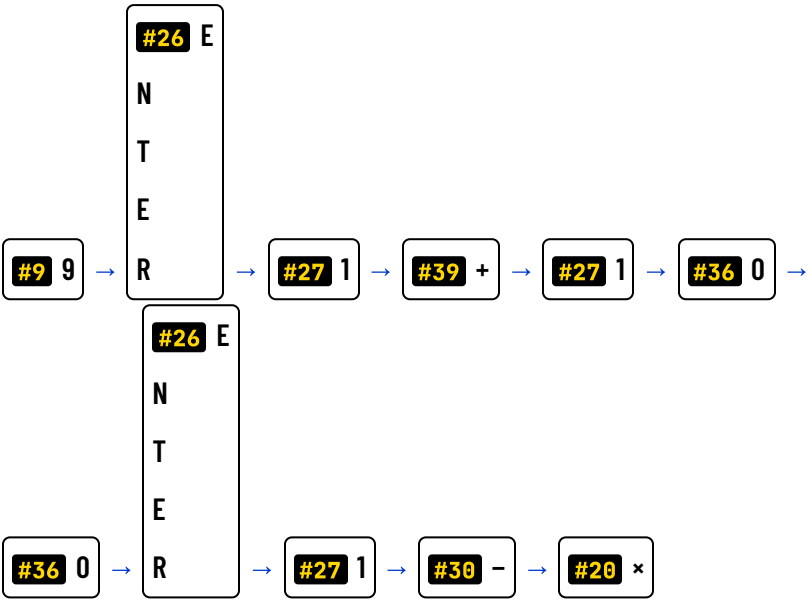
21. 12 (display 12.00)



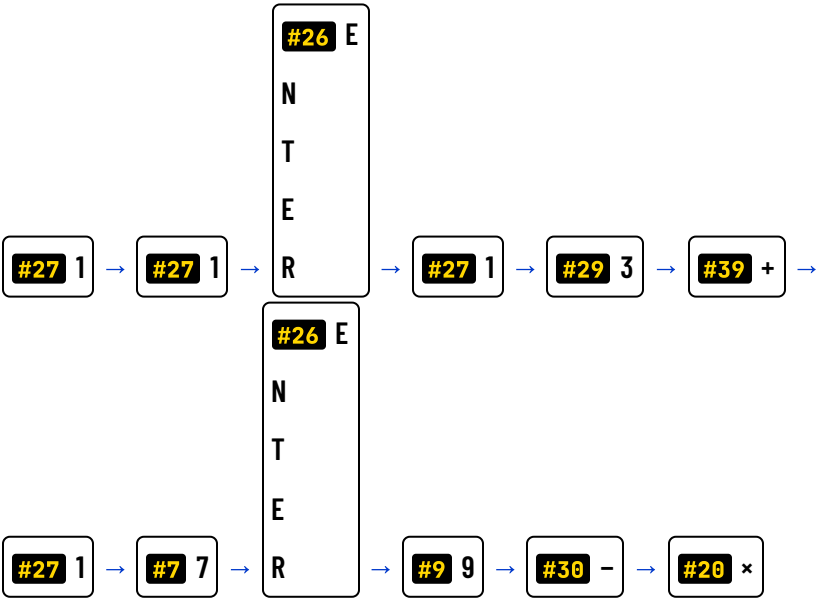
22. 500 (display 500.00)



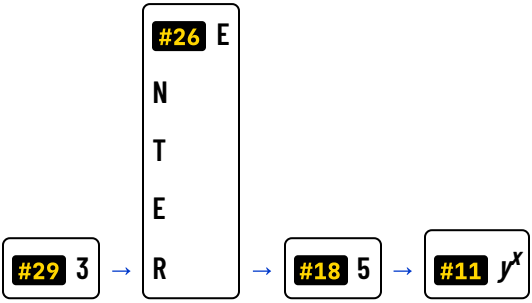
23. 990 (display 990.00)



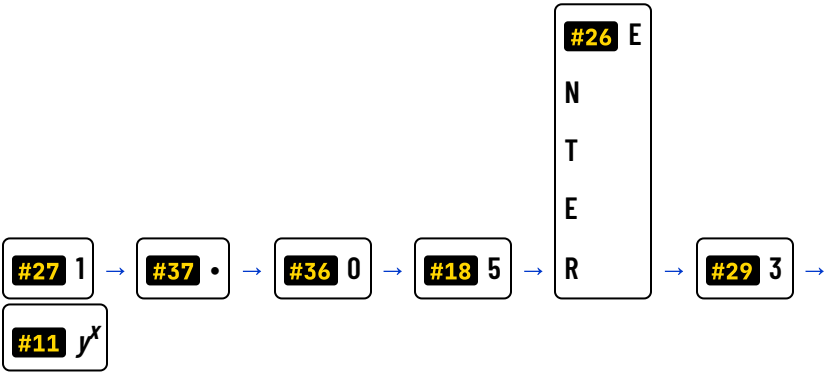
24. 192 (display 192.00)



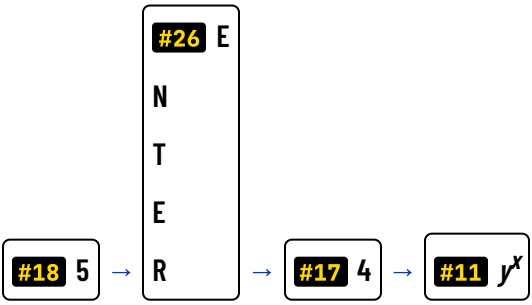
25. 243 (display 243.00)



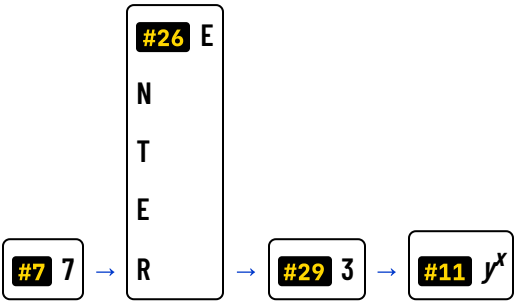
26. 1.157625 (display 1.16)



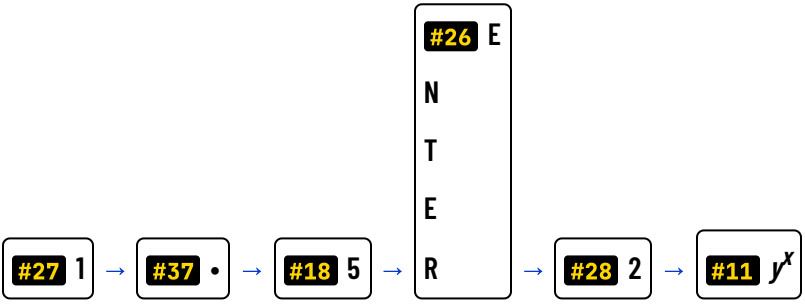
27. 625 (display 625.00)



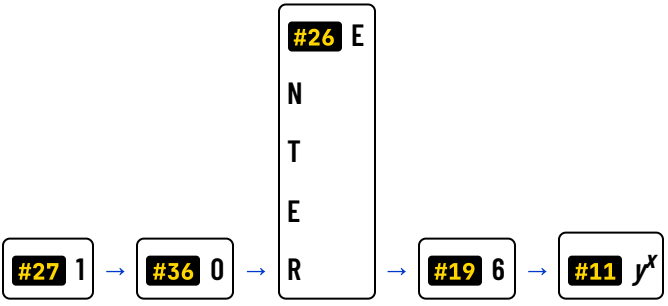
28. 343 (display 343.00)



29. 2.25 (display 2.25)



30. 1000000.000 (display 1,000,000.00)



31. 1.41421356 (display `1.41`)

#28 2 → #26 ENTER → #36 0 → #37 • → #18 5 → #11 y^x

32. 0.125 (display `0.13`)

#8 8 → #12 1/x

33. 4 (display `4.00`)

#36 0 → #37 • → #28 2 → #18 5 → #12 1/x

34. 0.0625 (display `0.06`)

#27 1 → #19 6 → #12 1/x

35. 0.2 (display `0.20`)

#18 5 → #12 1/x

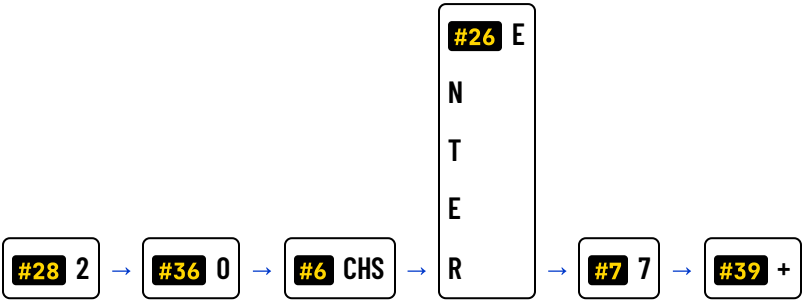
36. 0.025 (display `0.03`)

#17 4 → #36 0 → #12 1/x

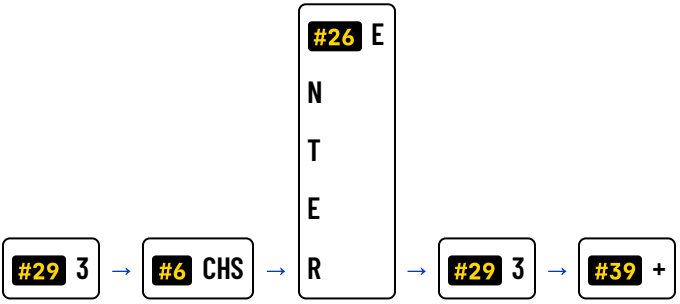
37. 5 (display `5.00`)

#36 0 → #37 • → #28 2 → #12 1/x

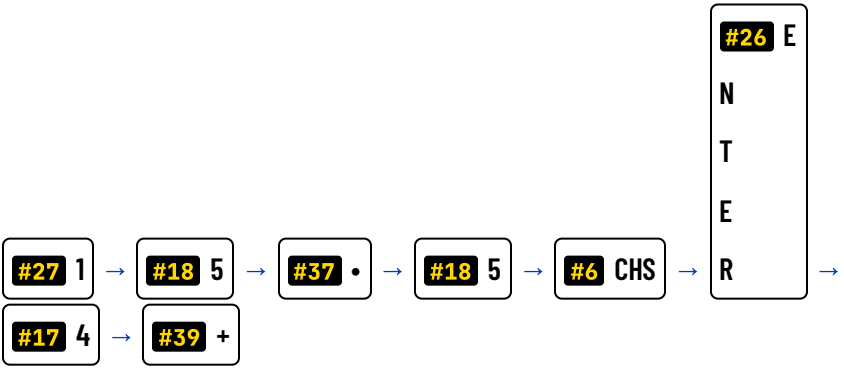
38. -13 (display -13.00)



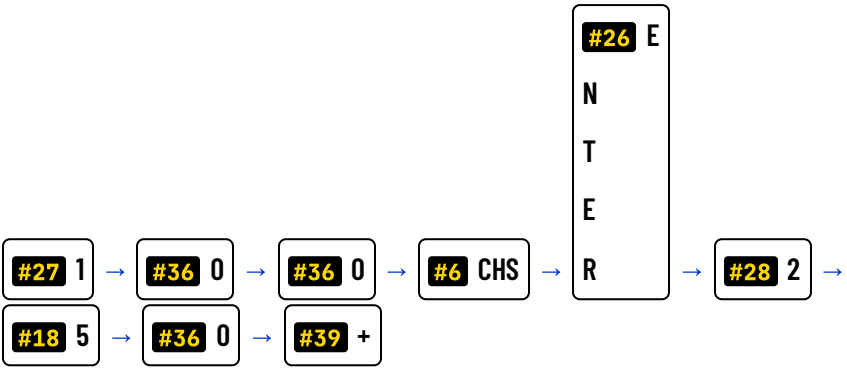
39. 0 (display 0.00)



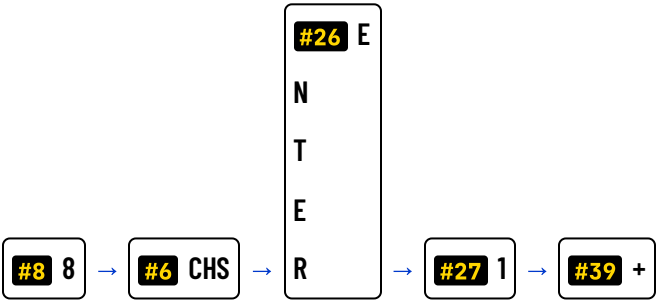
40. -11.5 (display -11.50)



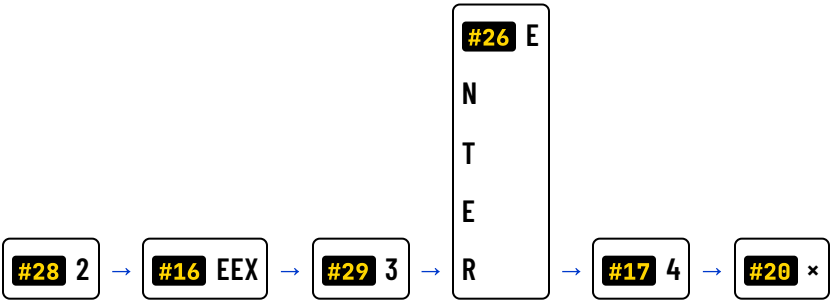
41. 150 (display 150.00)



42. -7 (display -7.00)



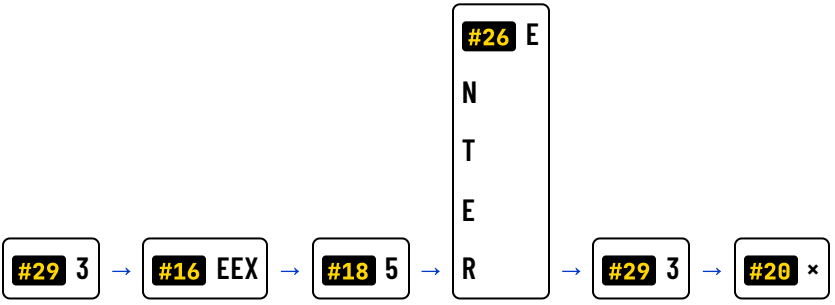
43. 8000 (display 8,000.00)



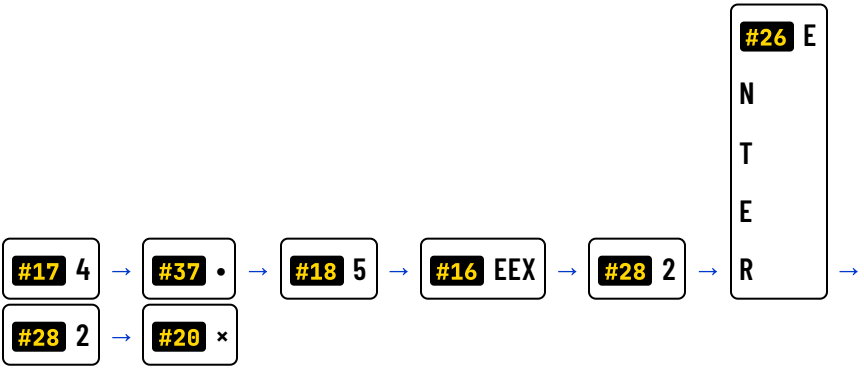
44. 30000 (display 30,000.00)



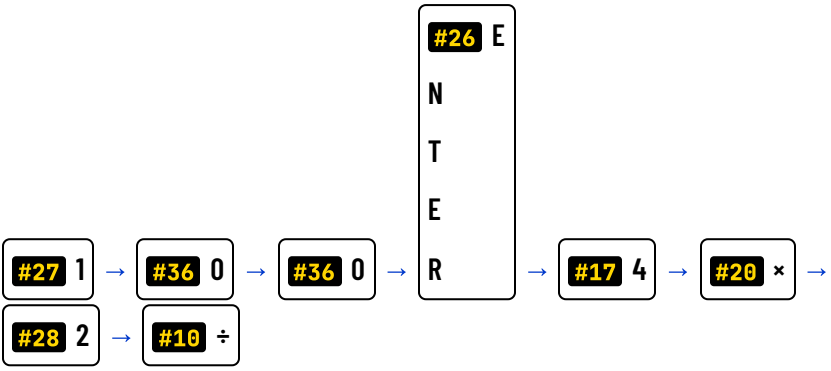
45. 900000 (display 900,000.00)



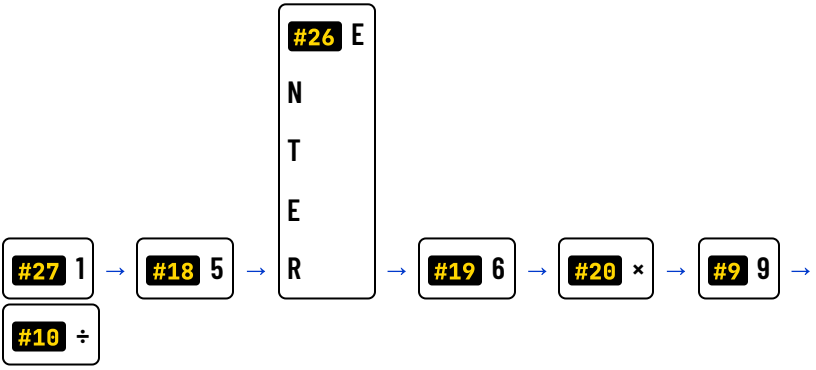
46. 900 (display 900.00)



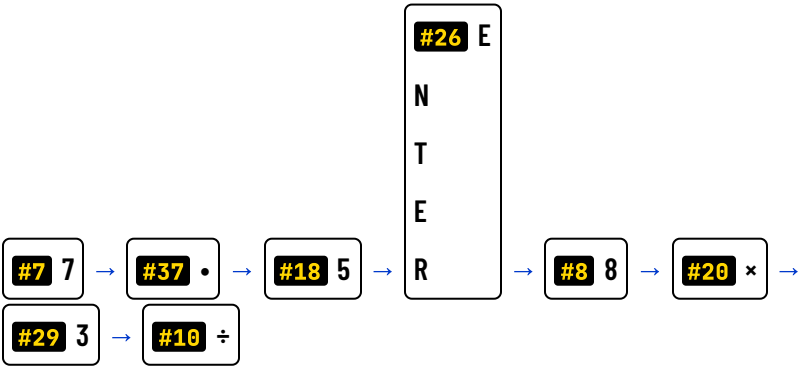
47. 200 (display 200.00)



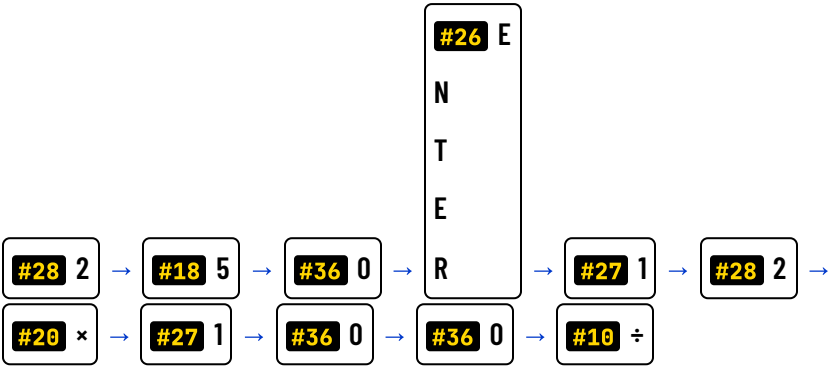
48. 10 (display 10.00)



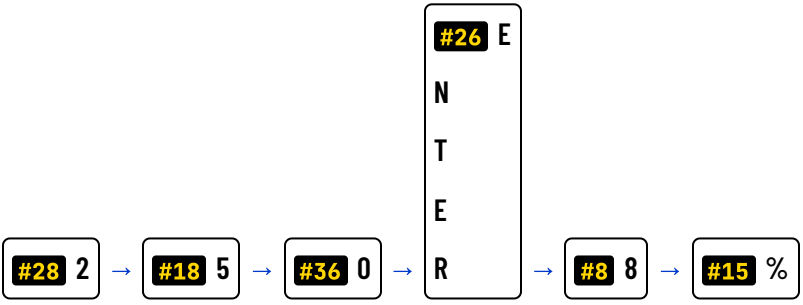
49. 20 (display 20.00)



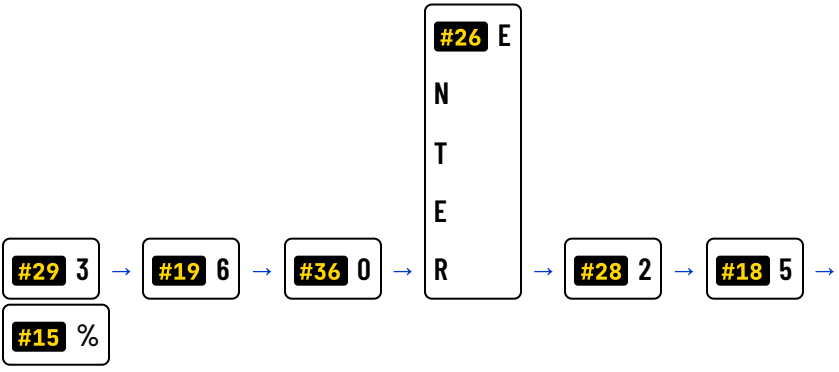
50. 30 (display 30.00)



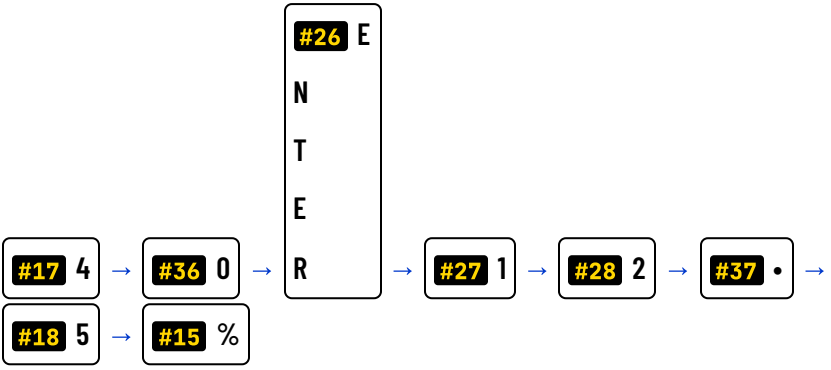
51. 20 (display 20.00)



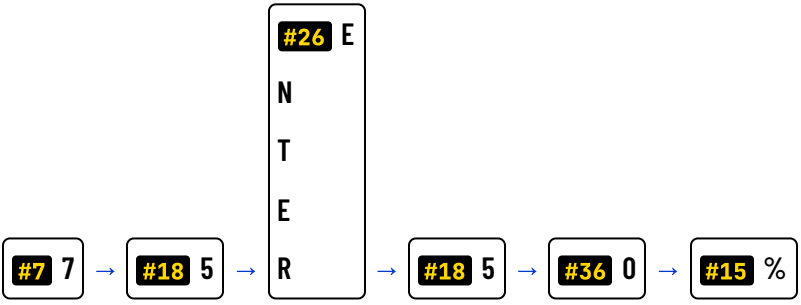
52. 90 (display 90.00)



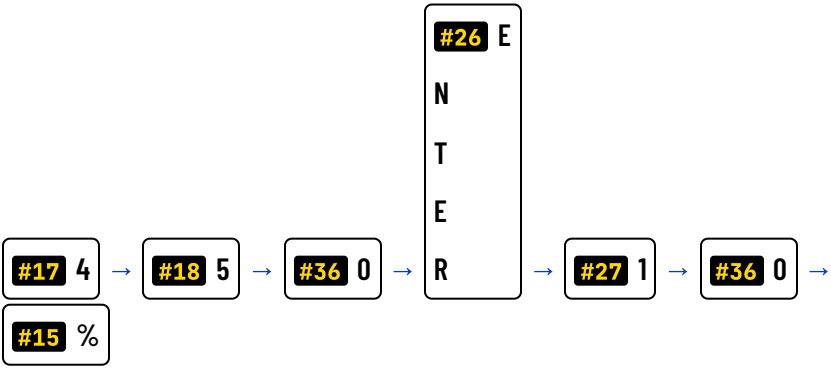
53. 5 (display 5.00)



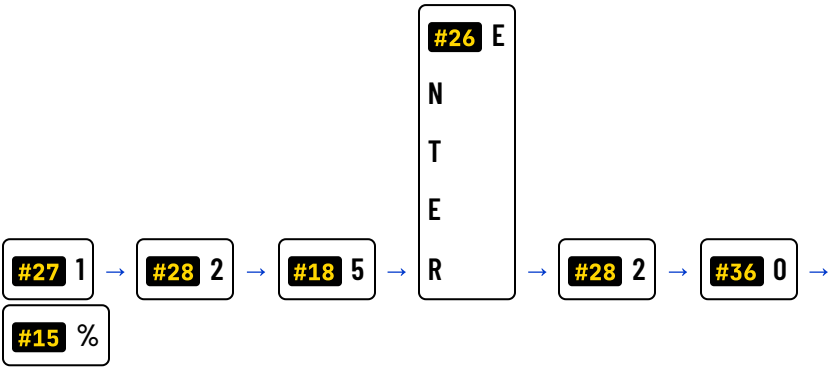
54. 37.5 (display 37.50)



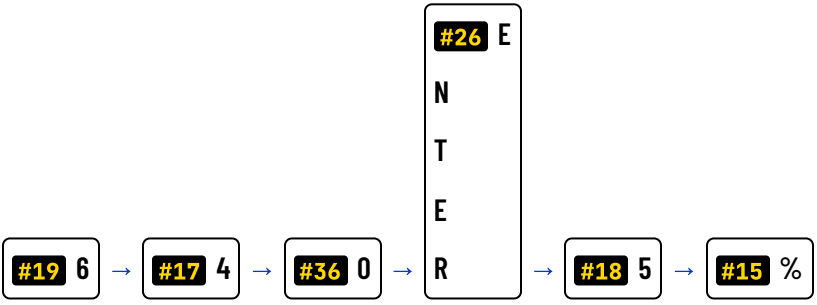
55. 45 (display 45.00)



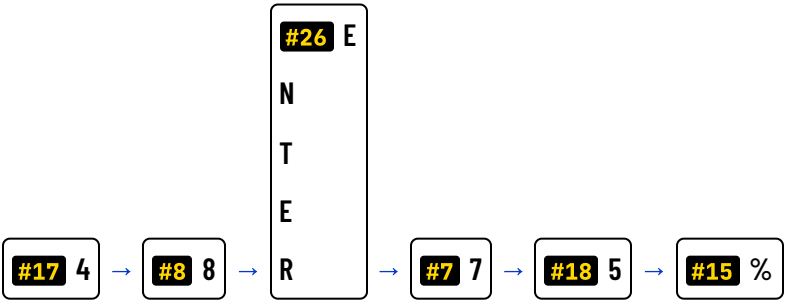
56. 25 (display 25.00)



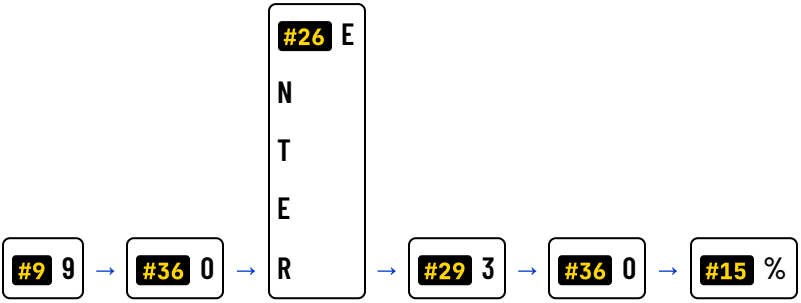
57. 32 (display 32.00)



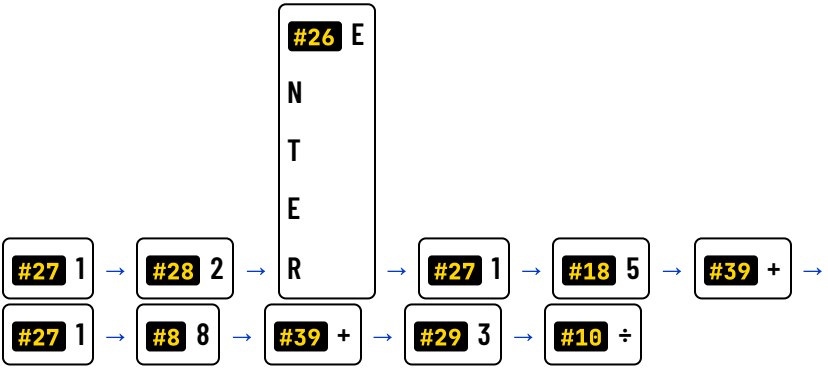
58. 36 (display 36.00)



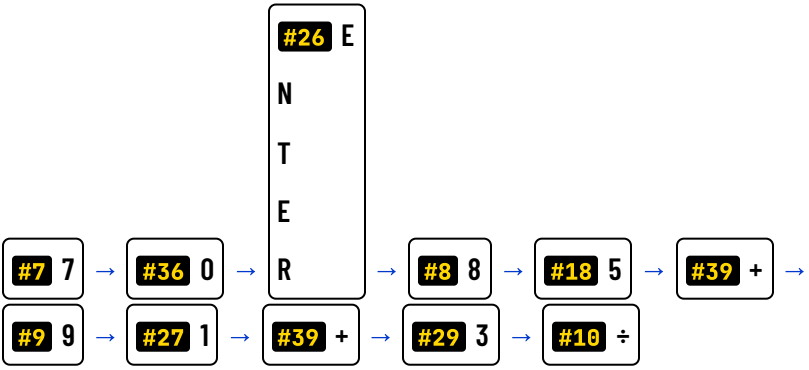
59. 27 (display 27.00)



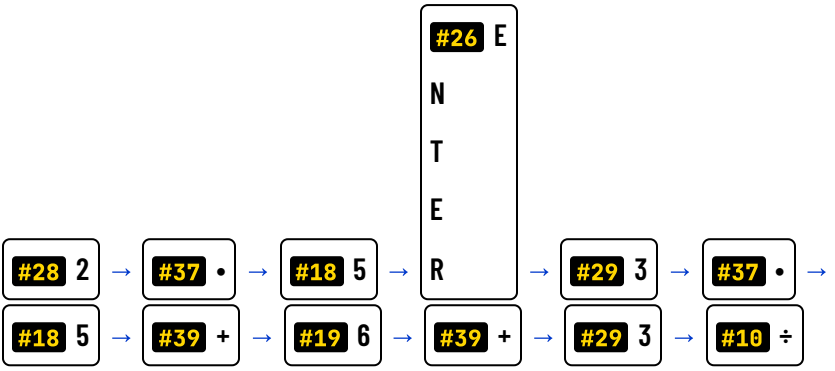
60. 15 (display 15.00)



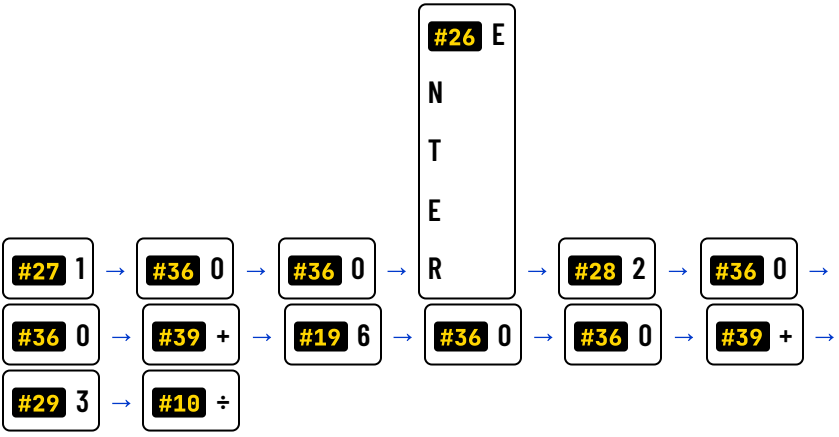
61. 82 (display 82.00)



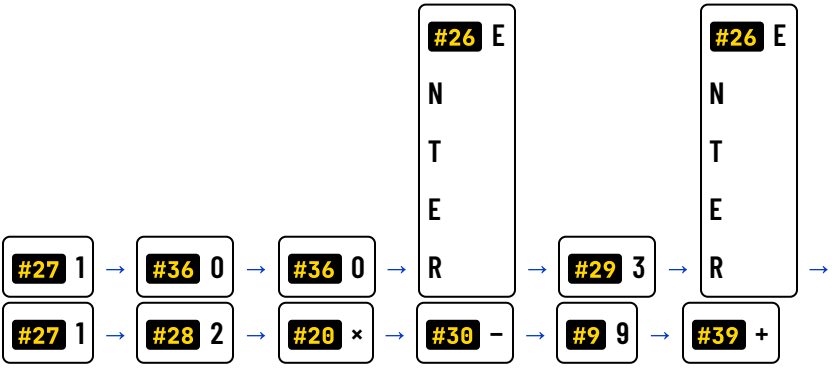
62. 4 (display 4.00)



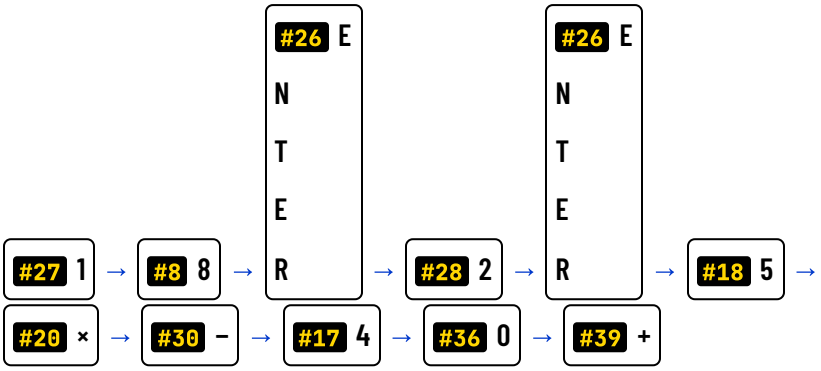
63. 300 (display 300.00)



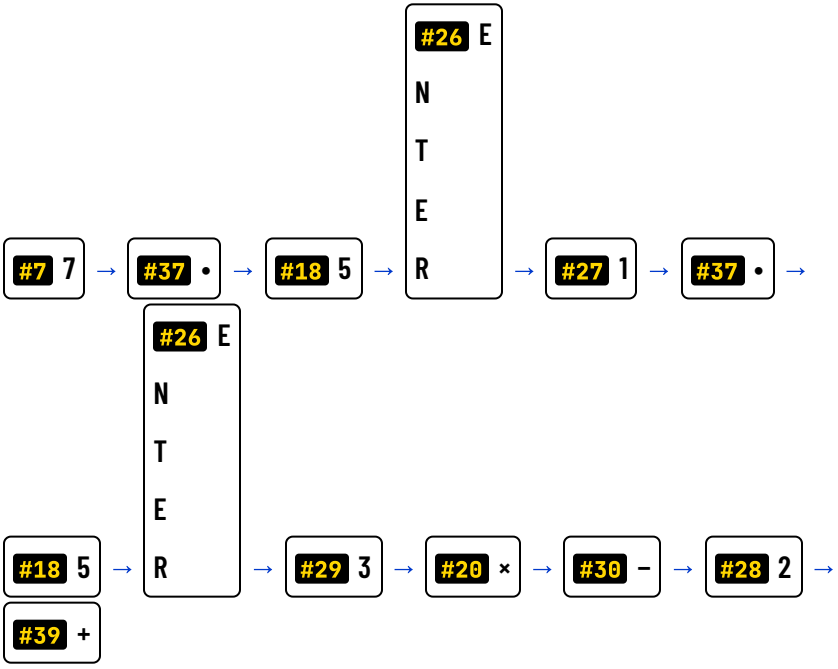
64. 73 (display 73.00)



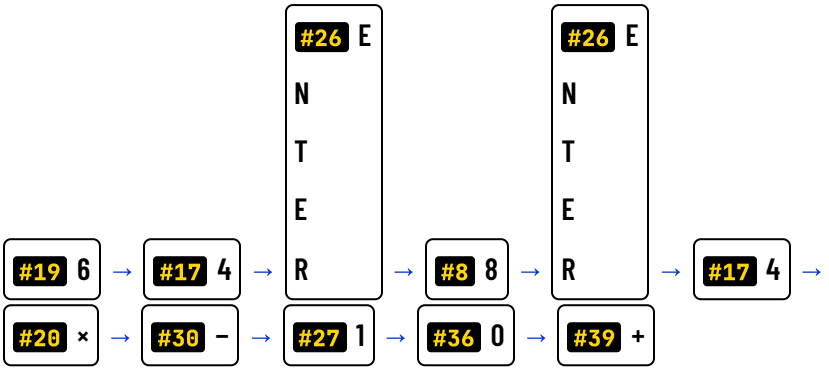
65. 48 (display 48.00)



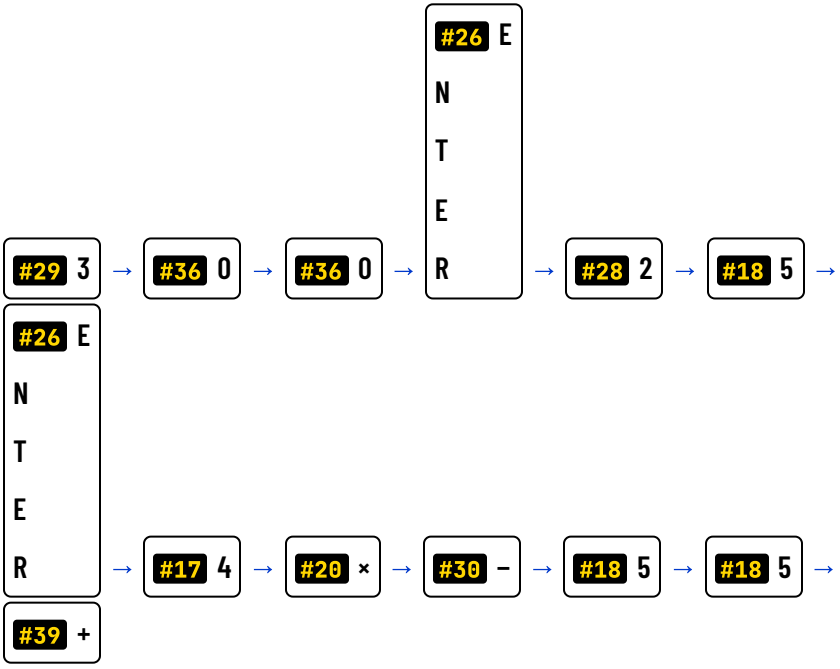
66. 5 (display 5.00)



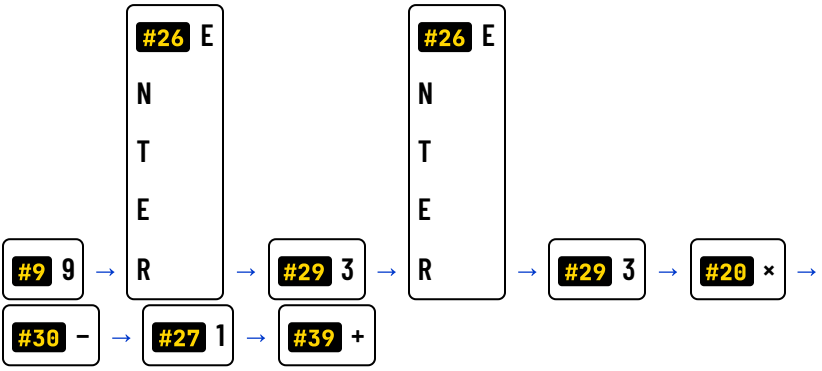
67. 42 (display 42.00)



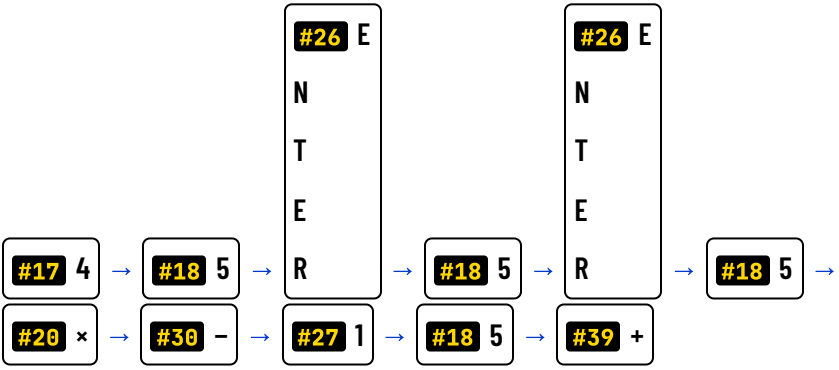
68. 255 (display 255.00)



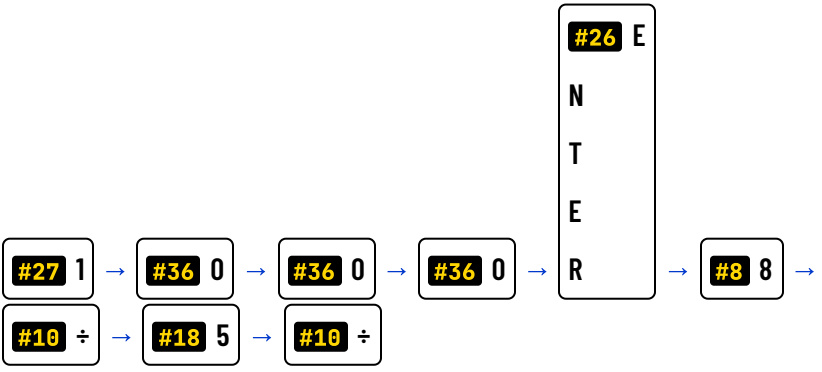
69. 1 (display 1.00)



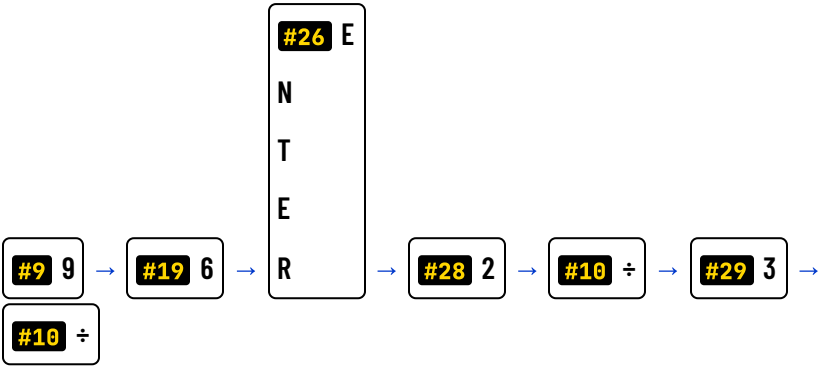
70. 35 (display 35.00)



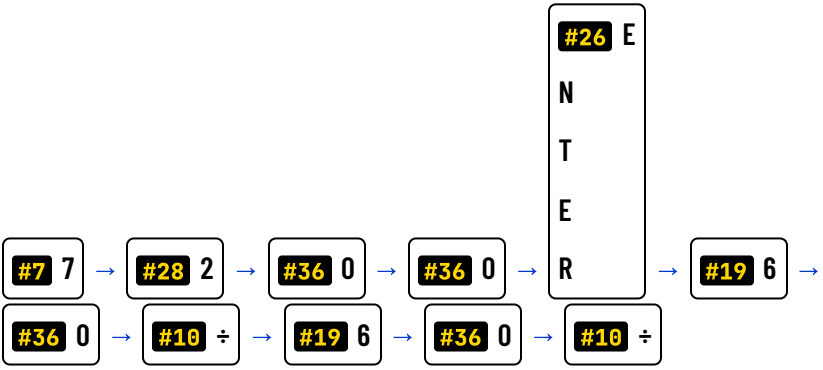
71. 25 (display 25.00)



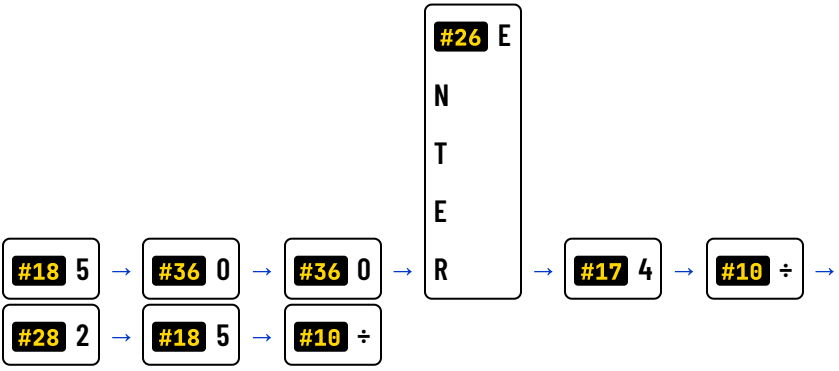
72. 16 (display 16.00)



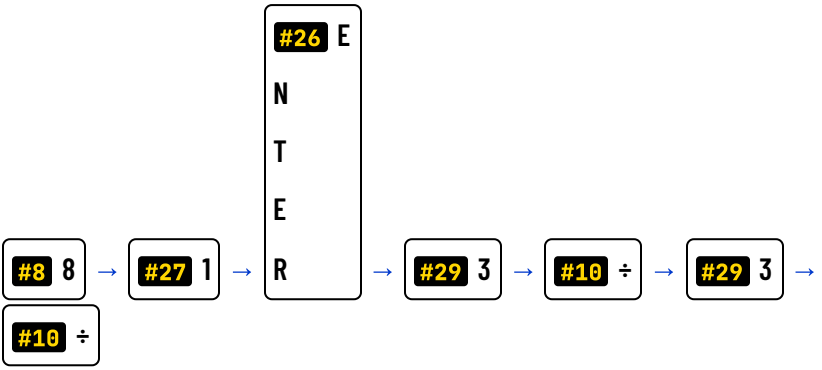
73. 2 (display 2.00)



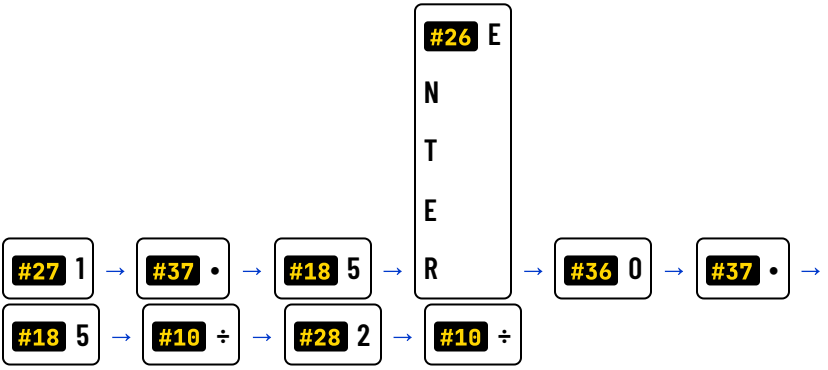
74. 5 (display 5.00)



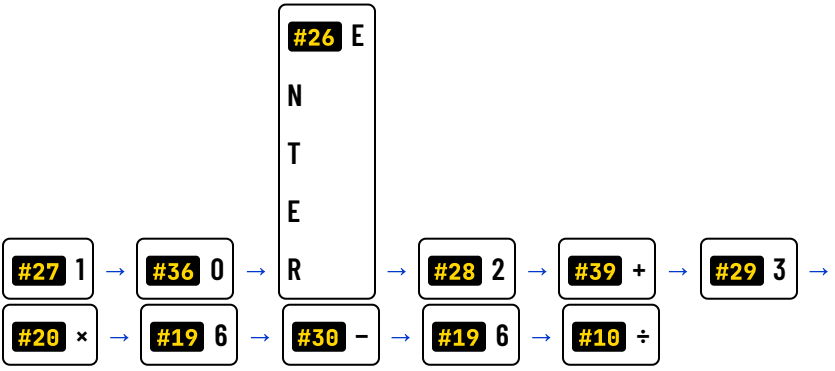
75. 9 (display 9.00)



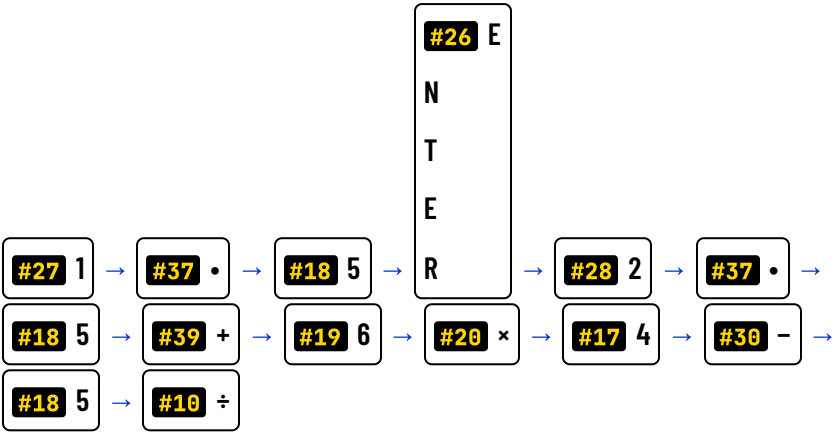
76. 1.5 (display 1.50)



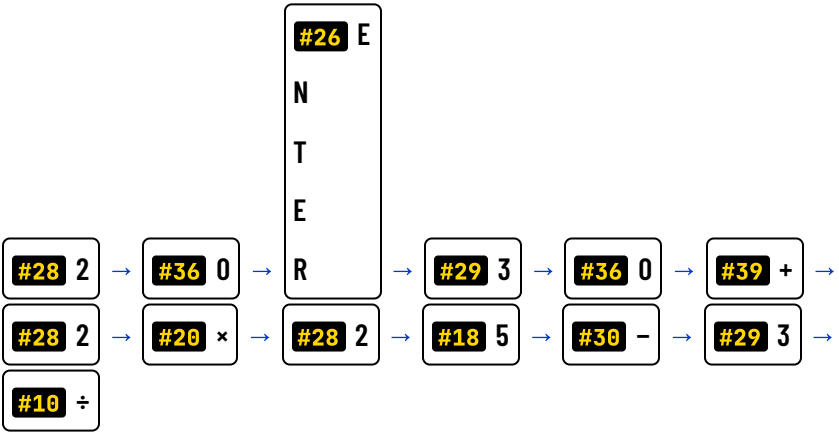
77. 5 (display 5.00)



78. 4 (display 4.00)



79. 25 (display 25.00)



80. 10 (display 10.00)

#7 7 → #26 ENTER → #8 8 → #39 + → #9 9 → #20 × → #27 1 → #18 5 → #30 − → #27 1 → #28 2 → #10 ÷

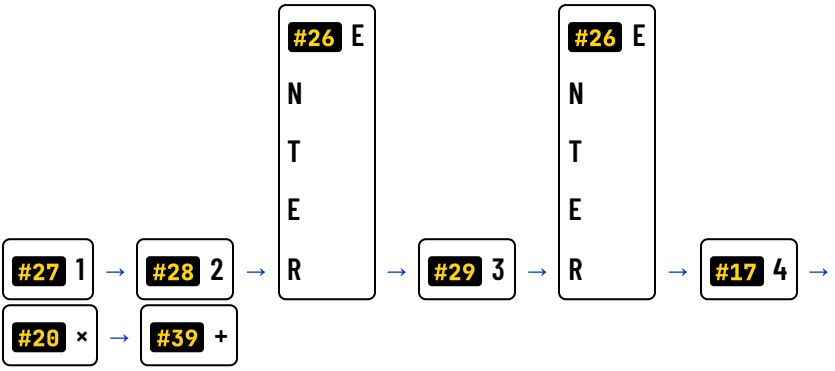
81. -466.08 (display -466.08)

#19 6 → #36 0 → #1 *n* → #17 4 → #37 • → #18 5 → #26 ENTER → #27 1 → #28 2 → #10 ÷ → #2 *i* → #28 2 → #18 5 → #36 0 → #36 0 → #36 0 → #3 PV → #36 0 → #5 FV → #4 PMT

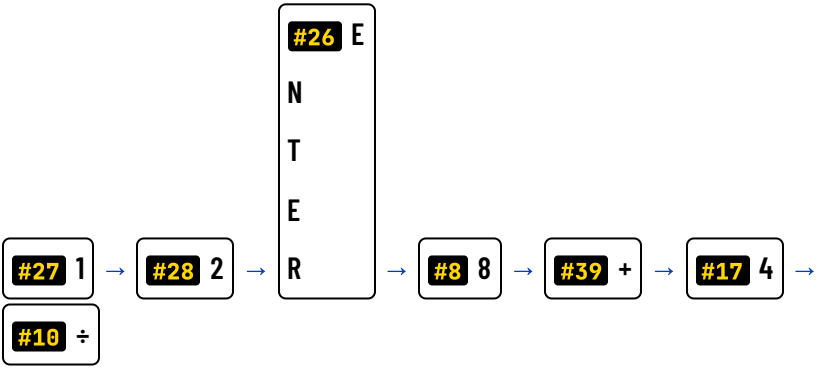
82. 9030.56 (display 9,030.56)

#28 2 → #36 0 → #1 *n* → #29 3 → #2 *i* → #18 5 → #36 0 → #36 0 → #36 0 → #6 CHS → #3 PV → #36 0 → #4 PMT → #5 FV

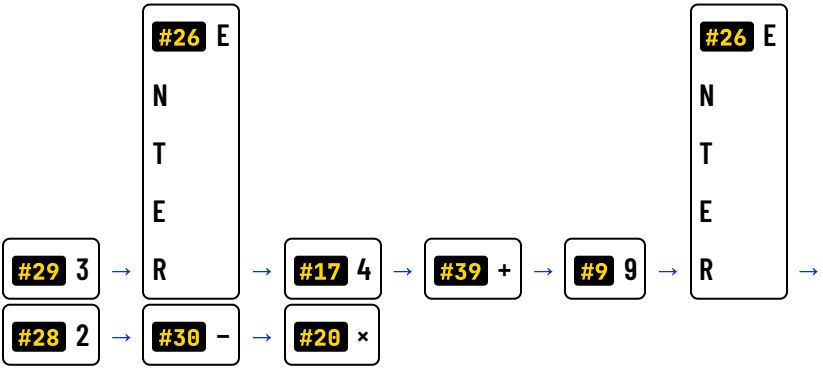
83. 24 (display 24.00)



84. 5 (display 5.00)



85. 49 (display 49.00)



86. 1024 (display 1,024.00)

#28 2 → #26 ENTER → #27 1 → #36 0 → #11 y^x

87. 0.25 (display 0.25)

#17 4 → #12 $1/x$

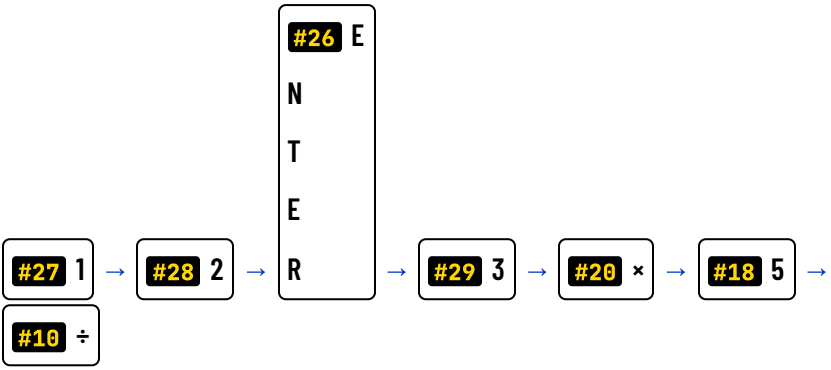
88. 7 (display 7.00)

#18 5 → #6 CHS → #26 ENTER → #27 1 → #28 2 → #39 +

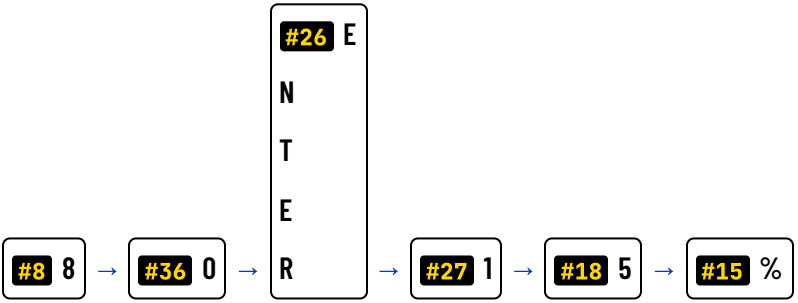
89. 1800 (display 1,800.00)

#19 6 → #16 EEX → #28 2 → #26 ENTER → #29 3 → #20 ×

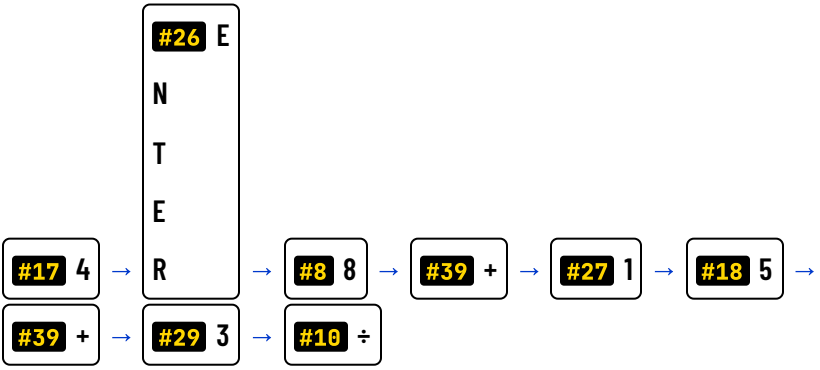
90. 7.2 (display 7.20)



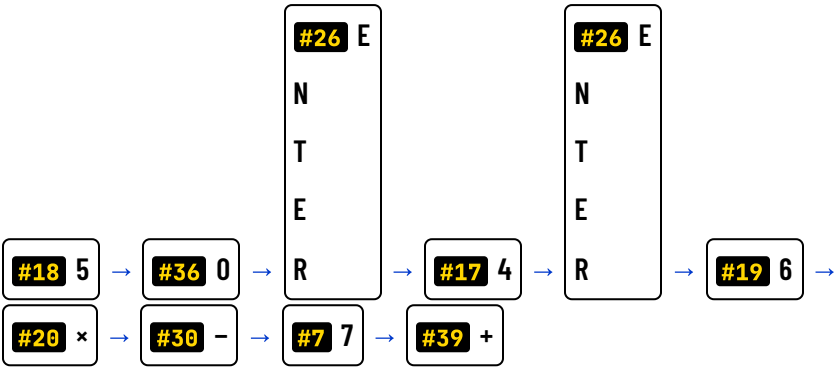
91. 12 (display 12.00)



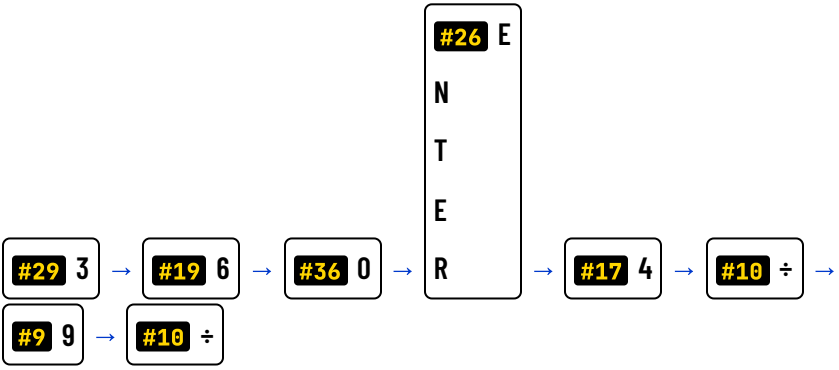
92. 9 (display 9.00)



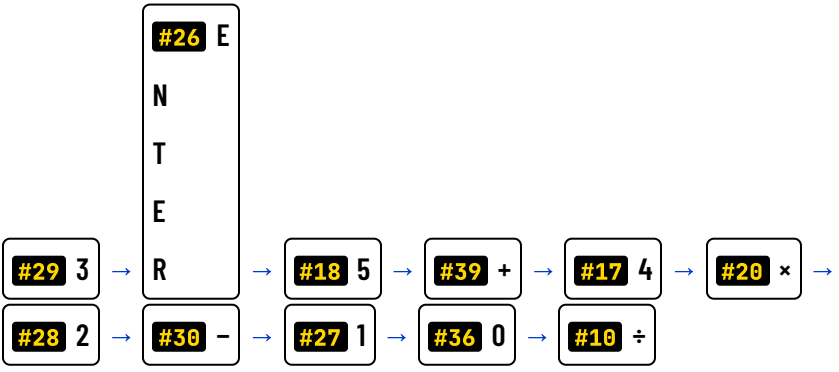
93. 33 (display 33.00)



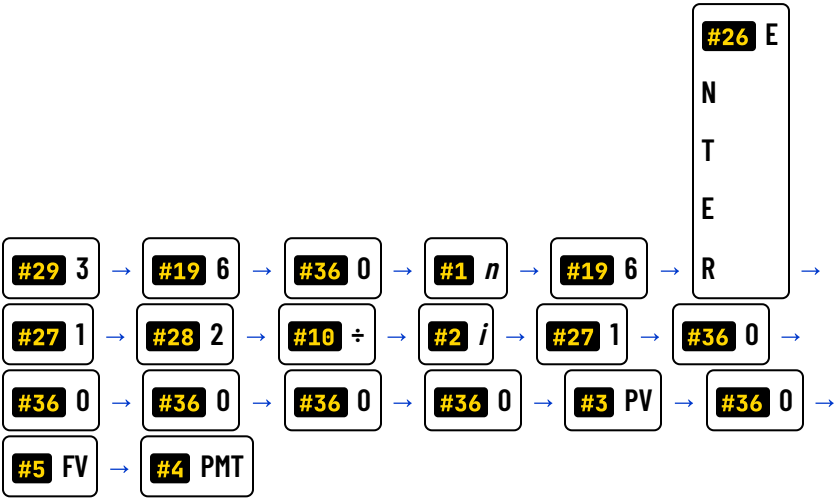
94. 10 (display 10.00)



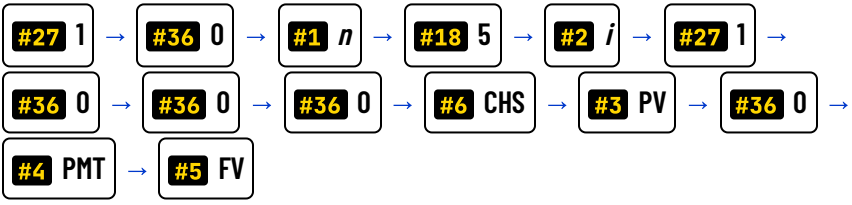
95. 3 (display 3.00)



96. -599.55 (display `-599.55`)



97. 1628.89 (display `1,628.89`)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #10
8. #11
9. #12
10. #13
11. #14
12. #15
13. #16
14. #20
15. #21
16. #22
17. #23
18. #24
19. #25
20. #26
21. #30
22. #31
23. #32
24. #33
25. #34
26. #35
27. #37
28. #38

29. #39
30. #32 f then #1 n
31. #32 f then #2 i
32. #32 f then #3 PV
33. #32 f then #4 PMT
34. #32 f then #5 FV
35. #32 f then #11 yx
36. #32 f then #12 1/x
37. #32 f then #13 %T
38. #32 f then #14 Δ%
39. #32 f then #15 %
40. #32 f then #21 R/S
41. #32 f then #22 SST
42. #32 f then #23 R↓
43. #32 f then #24 x≷y
44. #32 f then #25 CLx
45. #32 f then #26 ENTER
46. #32 f then #31 ON

Appendix B — Function finder (A–Z)

Function	Label	Keys
%T — percent of total	%T	#13
Addition +	+	#39
AMORT — amortization	AMORT	#32 → #1
Change sign (+/−) of the number being typed or the current result	CHS	#6
CLEAR FIN — clear the financial (TVM) registers	FIN	#32 → #24
CLEAR PREFIX — cancel a pending f / g / STO / RCL prefix	PREFIX	#32 → #26
CLEAR REG — clear all registers	REG	#32 → #25
CLEAR Σ — clear statistics registers	Σ	#32 → #22
CLx — clear X	CLx	#25
DB — declining-balance depreciation	DB	#32 → #15
Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored	•	#37
Digit 0	0	#36
Digit 1	1	#27
Digit 2	2	#28
Digit 3	3	#29
Digit 4	4	#17
Digit 5	5	#18
Digit 6	6	#19
Digit 7	7	#7
Digit 8	8	#8
Digit 9	9	#9

Division ÷	**÷**	**#10**
EEX — enter a power-of-ten exponent for the number in X	**EEX**	**#16**
ENTER — separate two numbers (push X to Y)	**ENTER**	**#26**
f — gold prefix: next key does its upper function	**f**	**#32**
FV — future value	**FV**	**#5**
g — blue prefix: next key does its lower function	**g**	**#33**
i — interest rate per period (TVM)	**i**	**#2**
INT — simple interest	**INT**	**#32 → #2**
IRR — internal rate of return	**IRR**	**#32 → #5**
Multiplication ×	**×**	**#20**
n — number of periods (TVM)	**n**	**#1**
NPV — net present value of entered cash flows	**NPV**	**#32 → #3**
ON/OFF — power toggle	**ON**	**#31**
ON/OFF — power toggle	**OFF**	**#32 → #31**
P/R — program mode	**P/R**	**#32 → #21**
Percent %	%	**#15**
PMT — payment	**PMT**	**#4**
PRICE — bond price	**PRICE**	**#32 → #11**
PV — present value	**PV**	**#3**
R/S — run/stop	**R/S**	**#21**
R↓ — roll the stack down	**R↓**	**#23**
RCL — recall a memory / variable	**RCL**	**#35**
Reciprocal x^{-1} (1/x)	**1/x**	**#12**
RND — round X to the displayed decimals	**RND**	**#32 → #4**

Shows the message “Programs n/a” (feature not provided in this app)	PRGM	#32 → #23
SL — straight-line depreciation	SL	#32 → #13
SOYD — sum-of-years depreciation	SOYD	#32 → #14
SST — single step	SST	#22
STO — store the value into a memory / variable (press the target next)	STO	#34
Subtraction –	–	#30
x≷y — exchange X and Y	x≷y	#24
YTM — bond yield	YTM	#32 → #12
y^x — raise Y to the power X (RPN: base first, then exponent)	yx	#11
Δ% — percent change	Δ%	#14
Σ+ — enter a statistics data point	Σ+	#38

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

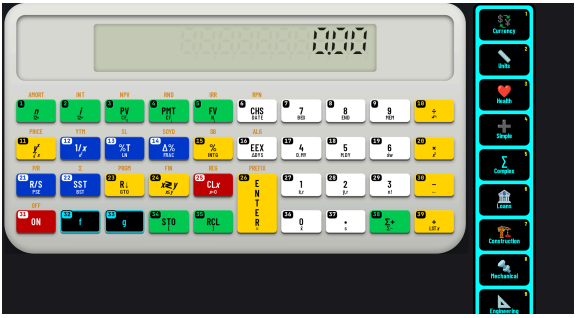
Landscape financial RPN. TVM with n, i, PV, PMT, FV (enter value + key; press the key alone to compute), g 12× / 12÷, BEG/END; cash flows CF0, CFj, Nj with NPV/IRR; AMORT, INT (simple interest), dates (D.MY/M.DY, DATE, ΔDYS), depreciation SL/SOYD/DB, statistics Σ+, x̄, s, ŷ,r, x̂,r, x̄w; f + digit sets the number of decimals. PRICE/YTM and programming are not simulated. f CLEAR Σ/FIN/REG/PREFIX are on SST, x≷y, CLx and ENTER. Some function labels on this photo were garbled; the standard labels for each key are used.

Planned (not yet working)

- PRICE / YTM bond functions
- Keystroke programming
- Continuous memory snapshot
- Amortisation export (CSV)
- Loan comparison
- Undo
- Speech read-out
- Colour-blind key theme

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 n ⇧ AMORT	#2 i ⇧ INT	#3 PV ⇧ NPV
#4 PMT ⇧ RND	#5 FV ⇧ IRR	#6 CHS ⇧ RPN
#7 7	#8 8	#9 9
#10 ÷	#11 yx ⇧ PRICE	#12 1/x ⇧ YTM
#13 %T ⇧ SL	#14 Δ% ⇧ SOYD	#15 % ⇧ DB
#16 EEX ⇧ ALG	#17 4	#18 5
#19 6	#20 ×	#21 R/S ⇧ P/R
#22 SST ⇧ Σ	#23 R↓ ⇧ PRGM	#24 x≷y ⇧ FIN
#25 CLx ⇧ REG	#26 ENTER ⇧ PREFIX	#27 1
#28 2	#29 3	#30 −
#31 ON ⇧ OFF	#32 f	#33 g
#34 STO	#35 RCL	#36 0
#37 •	#38 Σ+	#39 +

U-45 CALCULATORS · Calculator #45 · cut out and keep